

ARDMS Vascular Mock Registry Exam B24

Questions first. Answer key at the end. No feedback. Corrected complete version.

1. If the stenosis/blockage reduces the vessel lumen to 1/2 of the normal vessel area, the velocity of flow at the stenosis must _____ to maintain the flow rate

- A. quadruple
- B. double
- C. increase by 200%
- D. decrease by 50%

2. An abnormal Penile/Brachial Index will be:

- A. < 1.0
- B. > 1.0
- C. < 0.65
- D. > 0.65

3. Which of the following treatments would not be recommended for chronic peripheral arterial disease?

- A. Angioplasty
- B. Thrombolytic therapy
- C. Bypass graft
- D. Coumadin therapy

4. The diagnosis of AAA is made once the aorta AP or width dimension exceeds ____.

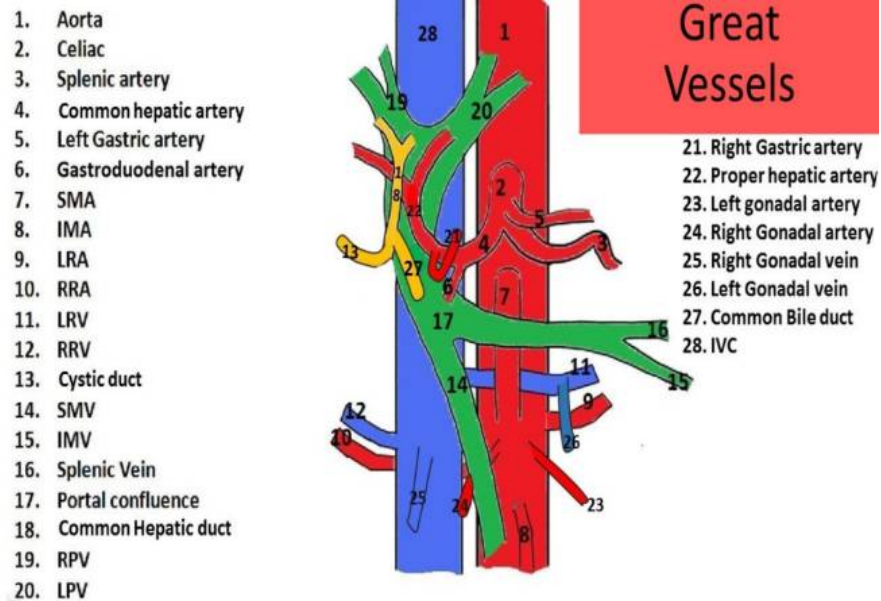
- A. 2.5cm
- B. 3.0cm
- C. 3.5cm
- D. 4.0cm

5. Which gonadal vein empties directly into the IVC?

- A. Right gonadal vein
- B. Left gonadal vein
- C. Right and left gonadal veins
- D. Neither gonadal vein empties directly into the IVC

6. Which of the following is a reason for lack of color fill in a transverse view of the distal femoral vein?

- A. scale settings too low
- B. the presence of reflux
- C. filter settings too low
- D. calcified plaque in the distal femoral artery



7. By dividing the pulsatility index proximal to a stenosis by the pulsatility index from distal to a stenosis, you can evaluate the:

- A. damping factor
- B. intraluminal diameter
- C. Reynolds number
- D. intraluminal area

8. What vein contains blood with a very high oxygen content?

- A. portal veins
- B. hepatic veins
- C. IVC
- D. pulmonary veins

9. Which of the following has no effect on the severity of an arterial stenosis?

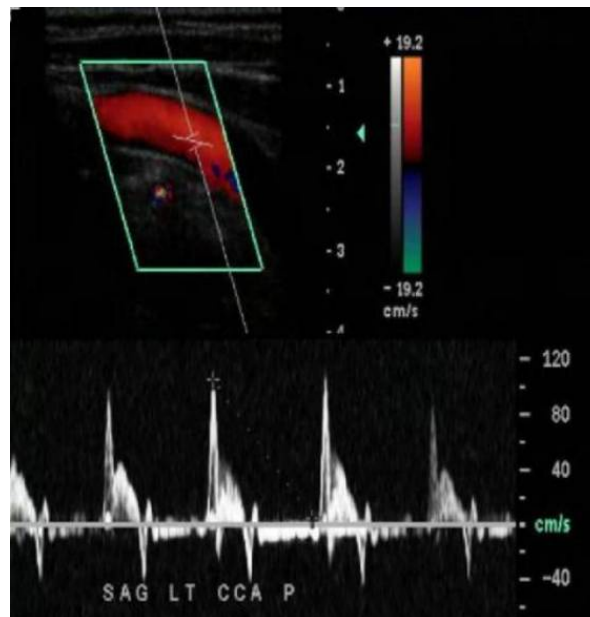
- A. vessel radius
- B. polycythemia vera
- C. valvular competence
- D. peripheral resistance

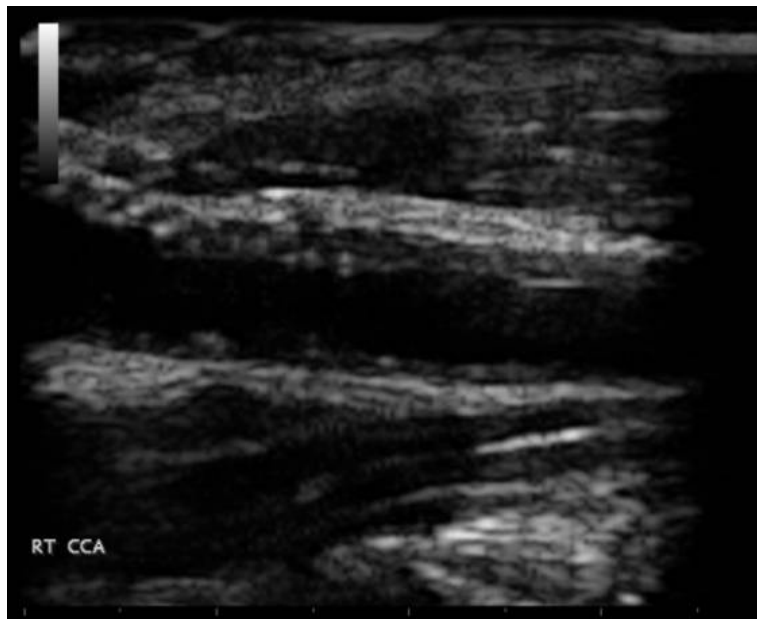
10. The PW Doppler tracing was obtained from the CCA. Which statement below is true regarding the image?

- A. Normal high resistance, triphasic flow is demonstrated
- B. Abnormal high resistance flow due to increased flow in Vertebral supply
- C. Abnormal high resistance flow due to ICA occlusion
- D. Abnormal high resistance flow due to proximal stenosis

11. Which of the following statements is true regarding the images displayed?

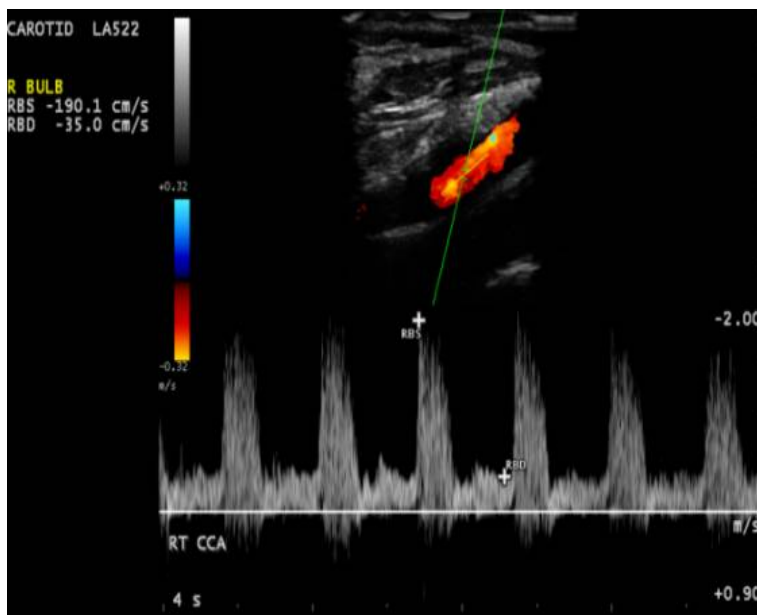
- A. The increase in PSV in the CCA is due to the multiple adjacent lesions causing increased resistance.
- B. There is focal atherosclerosis formation within the distal CCA.
- C. The increased PSV in the CCA is due to a single site of stenosis causing increased velocity and turbulence.
- D. The Doppler tracing indicates laminar flow.





12. What term refers to the phenomenon of a PW Doppler tracing displaying a wider distribution of the velocity of blood cells with varying degrees of arterial blockage?

- A. Spectral broadening
- B. Laminar flow
- C. Occlusion
- D. Turbulence



13. A patient presents for a carotid ultrasound. The exam demonstrates significant calcification in the proximal ICA on the left side and the Doppler signal is not obtained. Antegrade flow in the mid ICA has a peak velocity of 30cm/s and end diastolic velocity of 10 cm/s. How will you report these findings?

- A. Normal exam with limited evaluation of the proximal ICA.

- B. Suspected critical stenosis in the proximal left ICA, but the exam was limited by shadowing from atheroma formation.
- C. Critical stenosis proximal left ICA.
- D. Normal exam.

14. A double peaked waveform obtained during PPG evaluation of the digits of the left hand is indicative of:

- A. palmar arch stenosis
- B. Thoracic outlet syndrome
- C. Raynaud syndrome
- D. Subclavian stenosis

15. A common complication of cardiac catheterization is the formation of:

- A. Pseudoaneurysm
- B. Arterial dissection
- C. Fusiform aneurysm
- D. DVT

16. What setting on the photoplethysmography system should remain constant during an extremity arterial evaluation and if it is adjusted it must be documented on the tracing?

- A. Baseline level
- B. Size control
- C. Signal output
- D. Sweep speed

17. Which of the following is NOT an appropriate reason to terminate the treadmill exercise for a lower extremity arterial exam?

- A. patient complains of shortness of breath
- B. patient complains of leg pain
- C. patient complains of chest pain
- D. patient complains that the treadmill is too noisy

18. If an acute left ICA occlusion is present, which of the following statements is true?

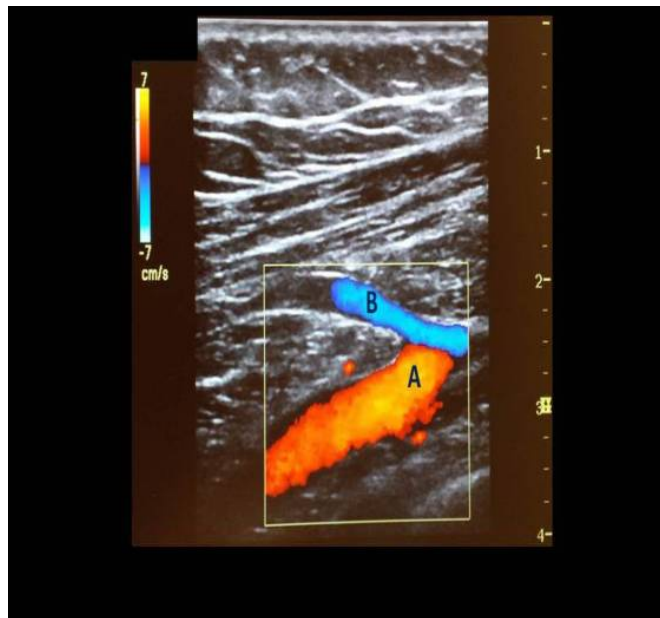
- A. The left MCA will demonstrate decreased or absent flow.
- B. The right MCA flow will be decreased or absent with normal flow in the right internal carotid siphon.
- C. The right ophthalmic artery will demonstrate decreased or absent flow
- D. The right MCA flow will be decreased or absent with flow reversal in the right internal carotid siphon.

19. Secondary Raynaud phenomenon:

- A. occurs in patients with an obstructed vascular system
- B. is a functional vasospastic disorder that occurs due to kidney disease
- C. occurs in patients with severe venous disease
- D. is a functional vasospastic disorder which occurs in patients with an unobstructed vascular system

20. Assuming proper transducer orientation, which of the following normal vessels could be displayed on the image?

- A. superior vena cava and right innominate vein
- B. left subclavian vein and cephalic vein
- C. left common femoral vein and greater saphenous vein
- D. right axillary vein and cephalic vein



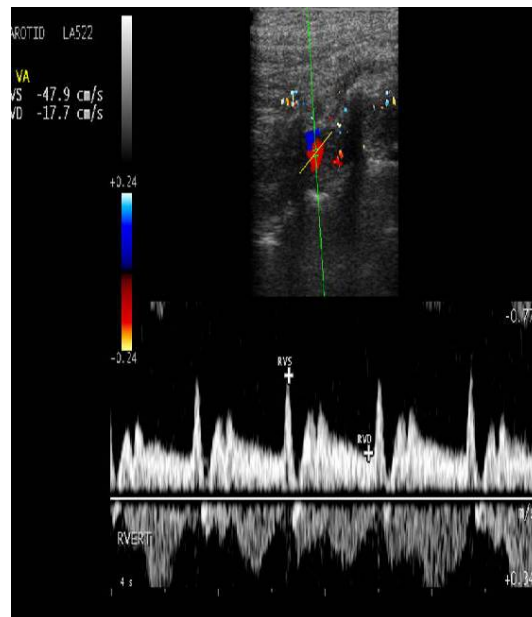
21. The image demonstrates the flow pattern taken in the right vertebral artery. What should be done next to better evaluate these findings?

- A. Have the patient perform the Valsalva maneuver and rescan during the strain
- B. Have the patient perform the Valsalva maneuver and rescan during the release
- C. Inflate a pneumatic cuff on the right arm to a level of 20mmHg above systolic pressure, wait 2 minutes to rapidly deflate and rescan
- D. Inflate a pneumatic cuff on the left arm to a level of 20mmHg above systolic pressure, wait 2 minutes to rapidly deflate and rescan

22. If the acceleration time is >140ms in both common femoral arteries:

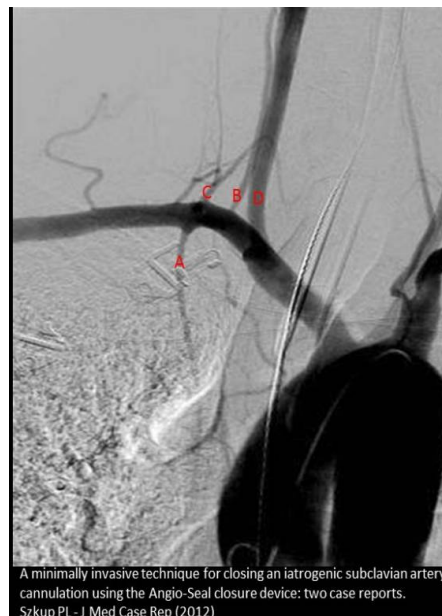
- A. bilateral internal iliac disease is suspected
- B. aortic disease is suspected

- C. unilateral external iliac disease is suspected
- D. flow is considered normal.



23. Letter A indicates which of the following structures?

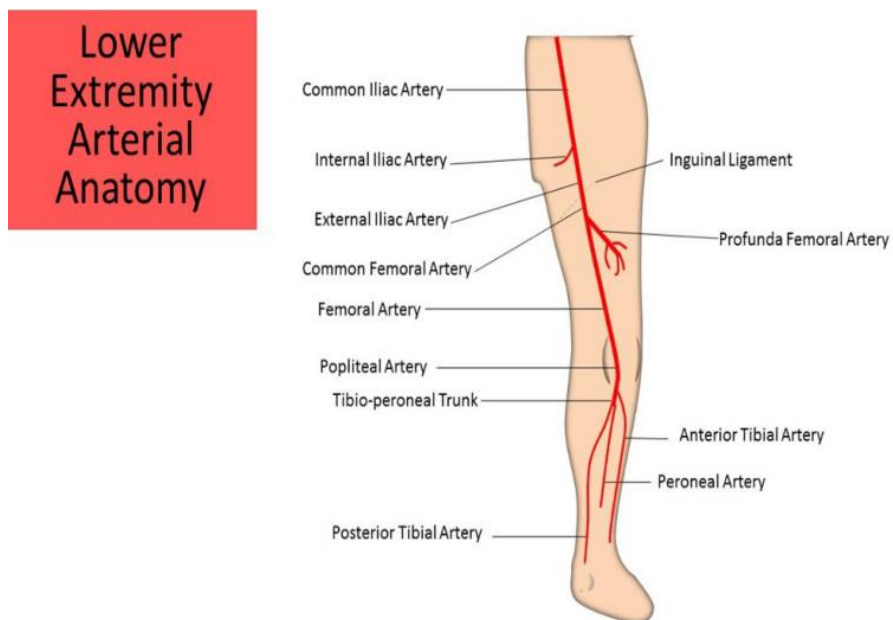
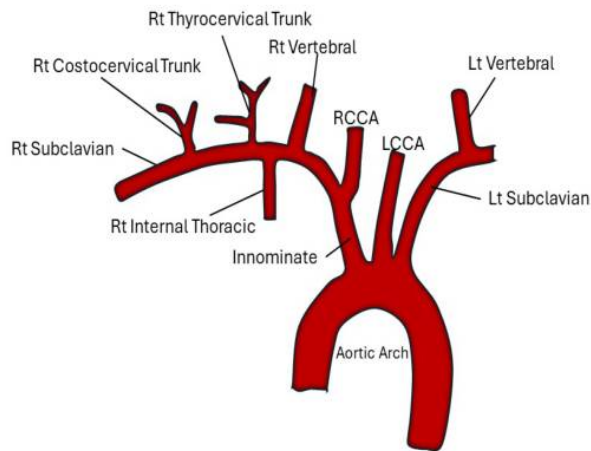
- A. Internal thoracic artery
- B. Superficial temporal artery
- C. Thyrocervical trunk
- D. Vertebral artery



24. If the tibioperoneal trunk is occluded, flow in which calf vessel(s) will demonstrate increased flow?

- A. anterior tibial artery

- B. posterior tibial artery
- C. peroneal artery
- D. posterior tibial artery and peroneal artery



25. This patient presents with a bruit, palpable mass, mild SOB and dizziness. These findings are most suggestive of:

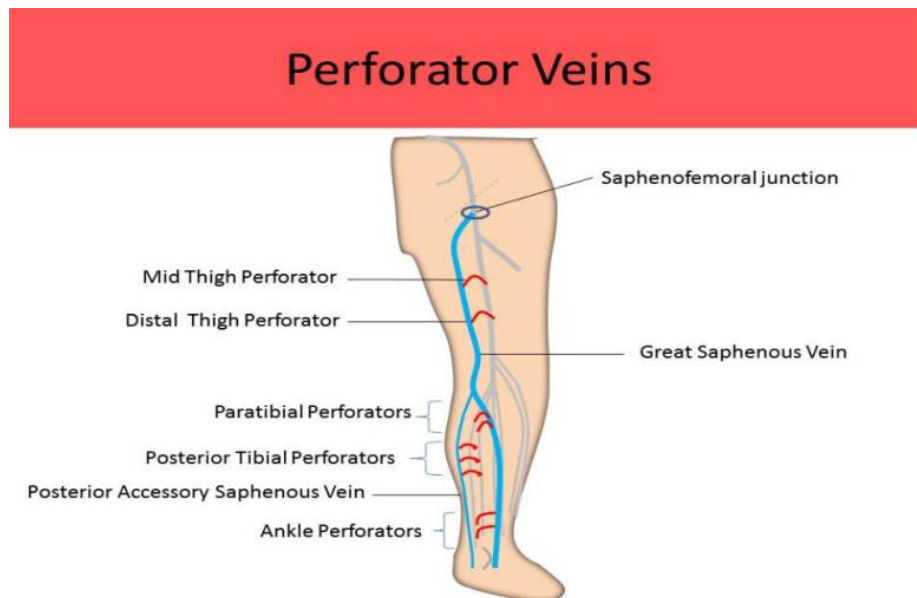
- A. AV fistula
- B. Pseudoaneurysm
- C. Carotid body tumor
- D. Thyroid mass

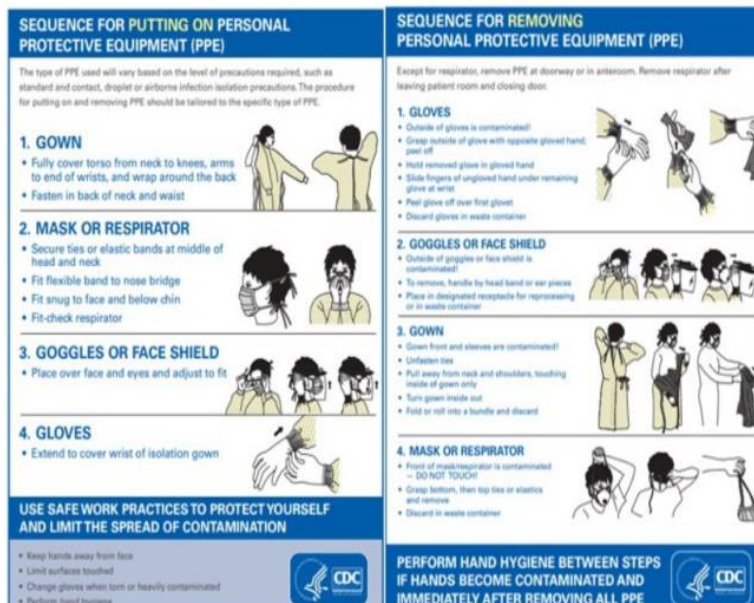
26. _____ carry(s) the blood from the GSV in the distal thigh into the distal femoral vein.

- A. Thigh perforators
- B. Posterior venous arch
- C. Lateral perforators
- D. Paratibial perforators

27. Which of the following indicates the proper order for putting on personal protective equipment?

- A. goggles, gown, mask, gloves
- B. gloves, gown, mask, goggles
- C. gown, mask, goggles, gloves
- D. mask, goggles, gloves, gown

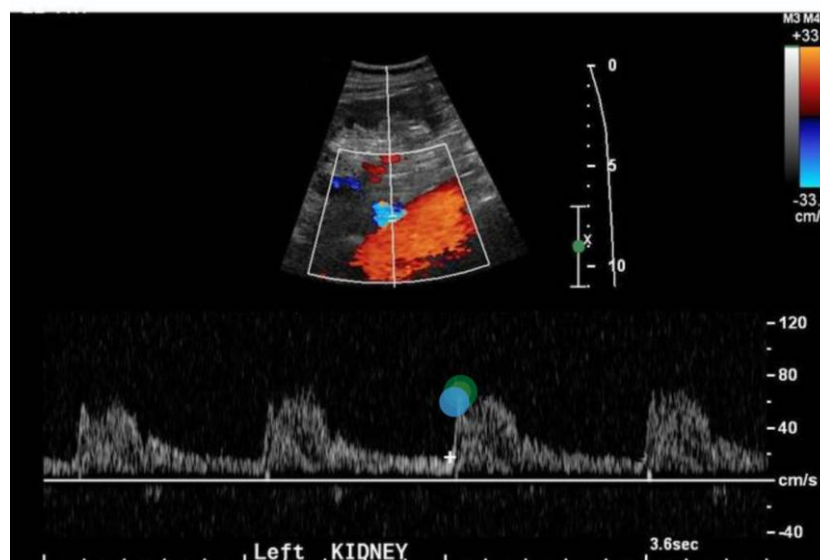




28. When calculating the ICA/CCA peak systolic velocity ratio,

- A. use the lowest CCA PSV and the highest ICA PSV obtained
- B. use the proximal CCA PSV and the ICA PSV obtained with the highest velocity
- C. use the distal CCA PSV and the ICA PSV obtained with the highest velocity
- D. use the proximal CCA PSV and the highest ICA PSV obtained from within the first 3cm of the vessel

29. Where should the second caliper be placed to measure the acceleration time of the renal artery waveform? Use your mouse to position the cursor and click to set your answer.



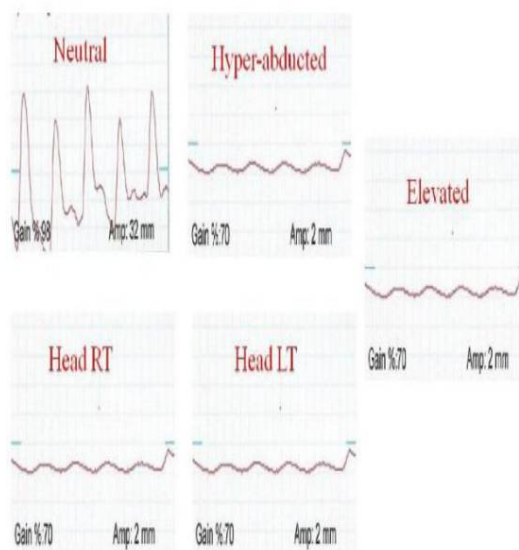
30. If new research shows that the minimum velocity criteria used to diagnose >60% renal artery stenosis is actually 225cm/s, not 180cm/s, how will this affect the accuracy of the

renal Doppler exam?

- A. decrease will vary with patient population
- B. no effect on the accuracy, only affects positive and negative predictive value
- C. increase will vary with patient population
- D. increase by 4.5%

31. The image demonstrates what type of vascular evaluation technique?

- A. Pulse Volume Recording
- B. Oculopneumoplethysmography
- C. Segmental pressures
- D. Analog Doppler



32. A patient presents for a 3-month follow-up on his aortic endograft. The aortic sac has increased in size by 0.8cm since the last ultrasound evaluation performed 2 months ago. No endoleak was detected on ultrasound or CT evaluation. What is the most likely cause for the diameter change?

- A. Reactive hyperemia
- B. May Thurner syndrome
- C. Graft migration
- D. Endotension

33. Which of the following Doppler information is necessary to calculate the resistive index?

- A. peak systolic velocity, average flow velocity and end diastolic velocity
- B. peak systolic velocity and peak diastolic velocity

- C. peak systolic velocity before and after a stenosis
- D. peak systolic velocity and end diastolic velocity

34. A patient presents with localized swelling and redness of the left calf that started two days ago. The referring physician requested a segmental pressure exam. What should you do first?

- A. Start with the brachial pressures and then perform the lower extremity pressures.
- B. Discuss the order with the radiologist to confirm that the exam that is ordered should be performed.
- C. Change the order to a venous Doppler after you speak to the patient regarding their symptoms.
- D. Start with the ankle pressures, then move up the leg and finish with the brachial pressures.

35. Where do the plantar arteries originate?

- A. from the anterior tibial artery anterior to the medial malleolus
- B. from the peroneal artery just past the Achilles tendon attachment at the heel
- C. from the anterior tibial artery posterior to the medial malleolus
- D. from the posterior tibial artery posterior to the medial malleolus

36. Normal brachial pressures should not differ more than _____ mmHg between arms.

- A. 30
- B. 20
- C. 10
- D. 5

37. Which of the following is a contraindication for a lower extremity segmental pressure exam?

- A. diabetes mellitus
- B. recent saphenous vein arterial bypass grafting
- C. ulceration on the 3rd and 4th toes
- D. current anticoagulant treatment

38. A patient is referred for an ultrasound of the aorta due to a history of Ehler-Danlos syndrome. What are you looking for on the exam?

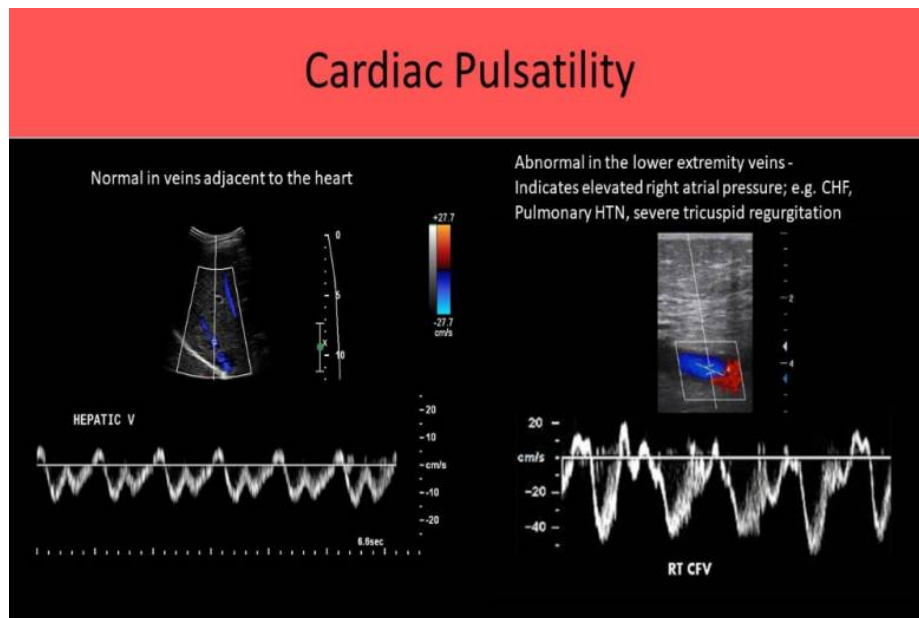
- A. Significant soft plaque formation in the distal aorta and proximal iliac arteries
- B. Absence of all anterior branches
- C. Significant fibrous tissue formation surrounding the distal aorta that is compressing the vessel
- D. Aneurysm

39. Which of the following terms would be used to describe abnormal venous flow in the deep veins of the thigh?

- A. cardiac pulsatility
- B. spontaneous
- C. phasic
- D. low velocity

40. Increased pressure in the venous system results in:

- A. decreased arterial pressure
- B. edema
- C. Baker cyst formation
- D. an increased risk of stroke



41. Which of the following would be the probe frequency of choice for a TCD exam?

- A. 2MHz
- B. 4MHz
- C. 7MHz
- D. 10MHz

42. Which vessel(s) supplies the majority of the blood to the hand?

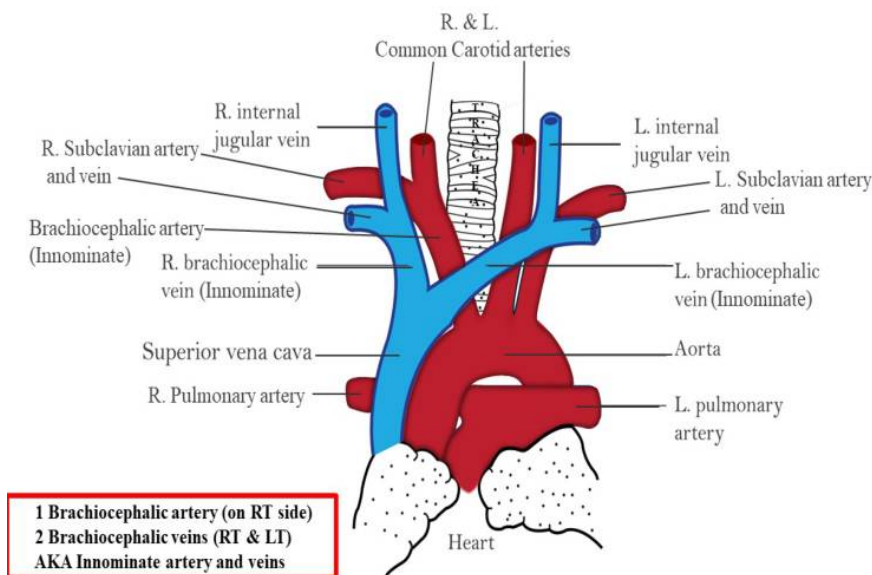
- A. radial artery
- B. ulnar artery
- C. metatarsal arteries
- D. digital veins

43. The left and right innominate veins merge to form the _____.

- A. Brachiocephalic vein
- B. inferior vena cava (IVC)
- C. Right atrium
- D. superior vena cava (SVC)

44. When continuous TCD monitoring is used to assist with a carotid endarterectomy, when is the TCD device first activated to begin monitoring the patient?

- A. immediately after the endarterectomy was performed but before the wound closure is performed
- B. within 1-2 hours after the endarterectomy was performed
- C. just before the surgery starts
- D. within 24hrs after the endarterectomy was performed



45. Which of the following is the best patient position for performing a transcranial Doppler exam to obtain a waveform from both of the middle cerebral arteries?

- A. seated on a stool, facing you and the machine
- B. supine with chin in neutral position
- C. seated on a stool, with their back to you and the machine
- D. left lateral decubitus with left hand placed under the head for support

46. A patient in the ER has just been diagnosed with acute DVT of the popliteal vein. What would be the physician's preferred treatment plan?

- A. Begin the use of compression stockings and perform a follow up exam in 30 days
- B. Insert a valvulotome into the popliteal vein and recanalize the lumen.

C. Immediate IV treatment with Heparin with 90 days of Warfarin medication once released from the hospital.

D. Perform sclerotherapy to resolve the clot.

47. The 10cm blood pressure cuffs are missing from the lab. You perform the 3-cuff method lower extremity segmental pressure exam with the 12cm and 14cm cuffs on a patient that is 5'5" 150lbs. What information that you will report will be invalid?

A. thigh pressure only

B. brachial pressure, calf pressure and ankle pressure

C. calf pressure only

D. brachial pressure only

48. Which of the following statements is true regarding the tunica intima?

A. An intimal dissection requires generation of a preliminary report and a stat review by a physician.

B. The vasa vasorum forms within the intimal layer.

C. The intimal layer of the artery will become less prominent with age and more difficult to visualize sonographically.

D. The tunica intima is responsible for holding the vessel shape and allowing for wall flexibility.

49. Vessel wall and plaque morphology are best evaluated in 2D when the beam is at what incident angle to the vessel?

A. 0 degrees

B. 45 degrees

C. 60 degrees

D. 90 degrees

50. Which of the following normal arteries is not palpable on physical exam?

A. dorsalis pedis

B. common femoral

C. peroneal

D. posterior tibial

51. A patient presents for an evaluation of a lower extremity synthetic bypass graft for suspected stenosis. What area of the graft should be evaluated first because it is the most common location of graft stenosis?

A. in the proximal graft

B. at the anastomosis sites

C. in the distal graft

D. native artery

52. Which of the following describes a cardiovascular application of the Bernoulli Principle?

- A. explains the reason for increased resistance to flow at an area of focal stenosis
- B. explains the reason for aliasing of the Doppler signal at high velocities
- C. explains the turbulence and layers of flow separation within the carotid bulb
- D. explains the relationship of vessel radius and flow volume

53. Which of the following Doppler imaging characteristics will not assist when trying to identify a vessel on a TCD exam?

- A. Sample size
- B. Direction of flow
- C. Sample location/depth
- D. Velocity of flow

54. A patient presents with intermittent left lower leg swelling for several months. The popliteal vein demonstrates a collapsed appearance with echogenic, thickened walls. These findings are most consistent with:

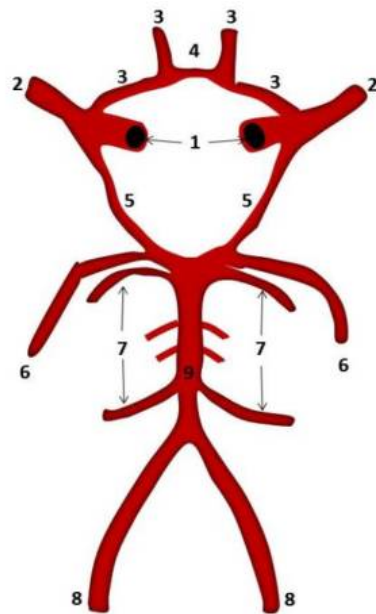
- A. acute femoral DVT
- B. chronic popliteal DVT
- C. vasculitis of the popliteal vein
- D. proximal venous stenosis

55. The posterior communicating artery connects the:

- A. right and left vertebral arteries
- B. right and left sides of the circle of willis
- C. anterior and posterior cerebral vessels
- D. ECA to the ICA

56. Which of the following is NOT normally a part of a test for vasculogenic impotence?

- A. Obtain Doppler samples of the CFA, PTA and DPA
- B. Obtain blood pressures in the arms and ankles
- C. Measure the length of the cavernous arteries
- D. Obtain Doppler samples of the cavernous arteries and veins in the penis



Circle of Willis

1. Internal carotid arteries (ICA)
2. Middle cerebral arteries (MCA)
3. Anterior cerebral arteries (ACA)
4. Anterior communicating artery
5. Posterior communicating artery
6. Posterior cerebral arteries (PCA)
7. Cerebellar arteries
8. Vertebral arteries
9. Basilar artery

57. If there is a 75% stenosis of the distal axillary artery, which of the following arteries will demonstrate a triphasic waveform?

- A. radial
- B. ulnar
- C. subclavian
- D. brachial

58. Which of the following is associated with CREST syndrome?

- A. Superficial phlebitis
- B. Primary Raynaud disease
- C. Secondary Raynaud phenomenon
- D. Aortic aneurysm

59. When performing a segmental pressure exam on a patient with claudication, which of the following describes the response to exercise in a patient with pseudoclaudication?

- A. Leg pain will not be associated with a drop in ankle pressure
- B. Leg pain will be associated with a drop in ankle pressure
- C. ABI will most likely be <0.5
- D. ABI will most likely be 0.5-0.9

60. Aneurysm formation affects the _____, while atherosclerotic disease affects the _____.

- A. extracranial arteries, intracranial arteries
- B. intimal layer of the vessel wall, media and adventitia layers of the vessel wall

- C. proximal arteries, distal arteries
- D. media and adventitia layers of the vessel wall, intimal layer of the vessel wall

61. Which of the following parameters is displayed as the x-axis on a Doppler tracing?

- A. motion
- B. velocity
- C. speed
- D. time

62. Which intracranial artery courses within the sylvian fissure?

- A. basilar artery
- B. middle cerebral artery
- C. posterior cerebral artery
- D. anterior cerebral artery

63. Which of the following is an expected finding with a positive evaluation for thoracic outlet syndrome?

- A. elevated blood pressure on the affected side with Adson's maneuver
- B. high amplitude PPG digital tracings on the affected side with arm adduction
- C. increased resistance in the proximal subclavian artery on the affected side with arm abduction
- D. increased flow velocity in the distal arm on the affected side with arm adduction

64. A diabetic patient presents for a segmental pressure exam. The calf and ankle vessels are non-compressible with up to 220mmHg. What alternative exam can be performed to evaluate lower extremity pressures?

- A. photoplethysmography with ankle cuffs
- B. Doppler ultrasound
- C. photoplethysmography and toe cuffs
- D. pulse volume recording exam

65. Which of the following requires the use of sterile technique?

- A. Laser ablation for telangiectasia
- B. Sclerotherapy
- C. Radiofrequency venous ablation
- D. Vein mapping

66. Which of the following is a branch of the ICA?

- A. Superior thyroidal

- B. Lingual
- C. Ophthalmic
- D. Temporal

67. The most common location of atherosclerotic formation in the cerebrovascular system?

- A. circle of Willis
- B. vertebrobasilar system
- C. ophthalmic and orbital arteries
- D. carotid bifurcation

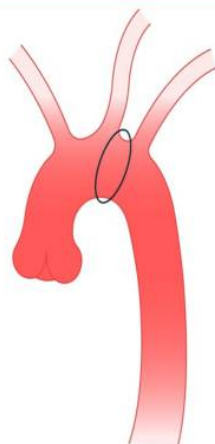
68. Which of the following conditions would cause an increase in the systemic BP in just the right arm?

- A. aortic coarctation just distal to the origin of the left common carotid artery
- B. thoracic outlet syndrome of the right arm
- C. right subclavian steal
- D. right subclavian artery stenosis

69. The Doppler tracing is most consistent with a tracing obtained from which normal extracranial vessel?

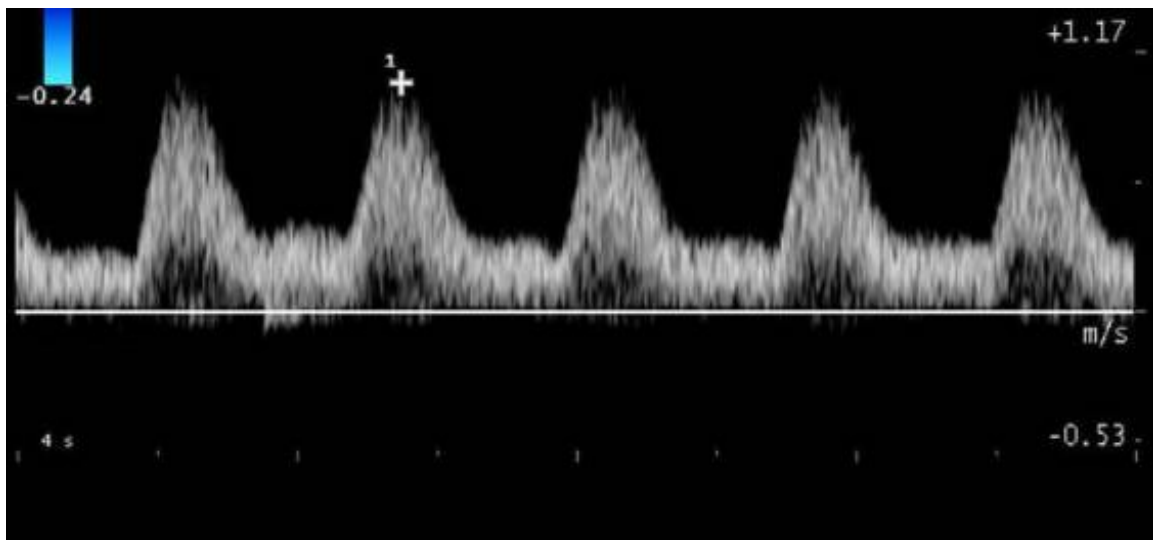
- A. external carotid artery
- B. common carotid artery
- C. internal carotid artery
- D. middle cerebral artery

Coarctation of the Aorta



If coarctation occurs just distal to the left CCA origin, the subclavian artery will be affected. Flow to the left arm will be diminished and the systemic BP in that arm will also be reduced

***Most common location of coarctation is just distal to the origin of the left subclavian artery, this causes elevated brachial pressures bilaterally and diminished bilateral ankle pressures.

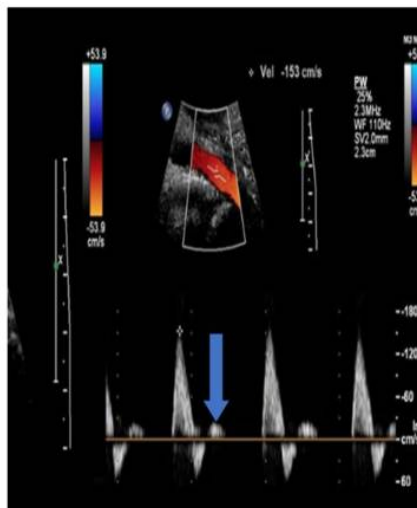


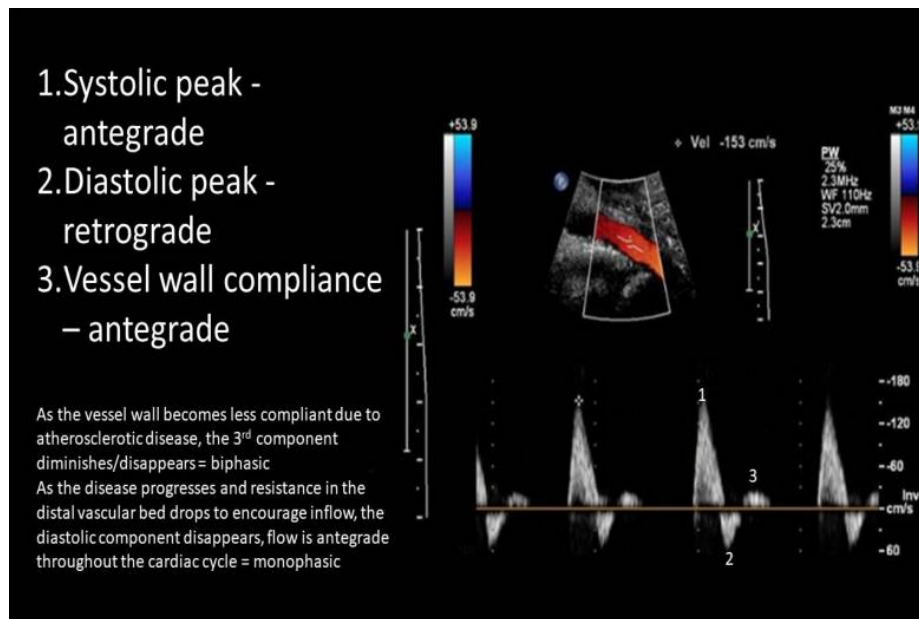
70. What causes the 3rd phase of flow in a normal extremity arterial waveform?

- A. cardiac systole
- B. relaxation of the left ventricle
- C. relaxation of the extremity musculature
- D. arterial wall compliance

71. The flow demonstrated below the baseline is:

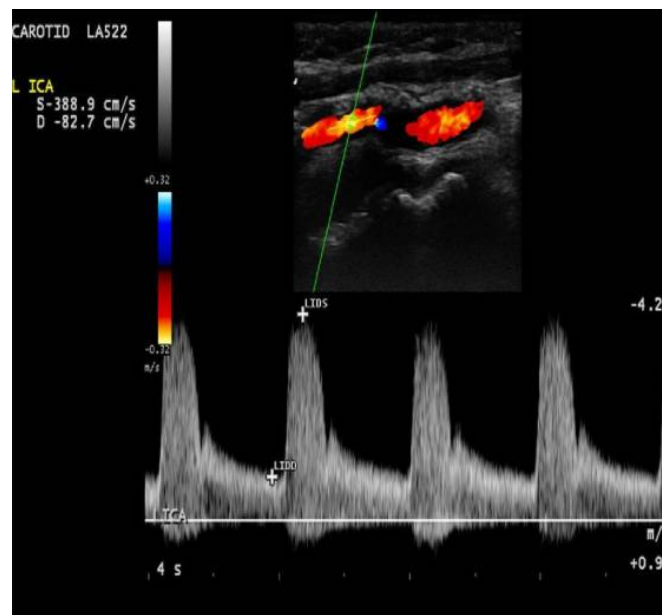
- A. caused by incoming collateral flow due to the significant stenosis.
- B. caused by turbulent flow at the site of the stenosis.
- C. noise caused by patient respiration.
- D. an example of cross talk artifact.

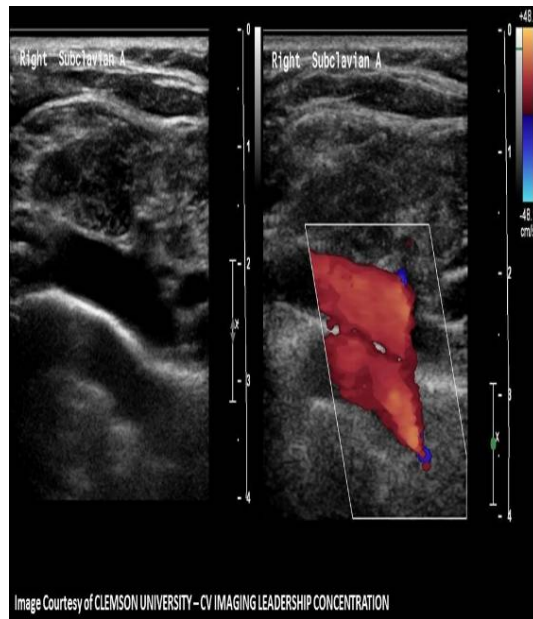




72. Which of the following is the most likely explanation for findings displayed on the image?

- A. The normal subclavian artery is displayed with significant reflux demonstrated in the subclavian vein
- B. The subclavian artery is duplicated and both vessels demonstrate normal flow
- C. The deeper vessel is a mirror image artifact caused by the apex of the lung
- D. The superficial vessel is a mirror image artifact caused by the apex of the lung



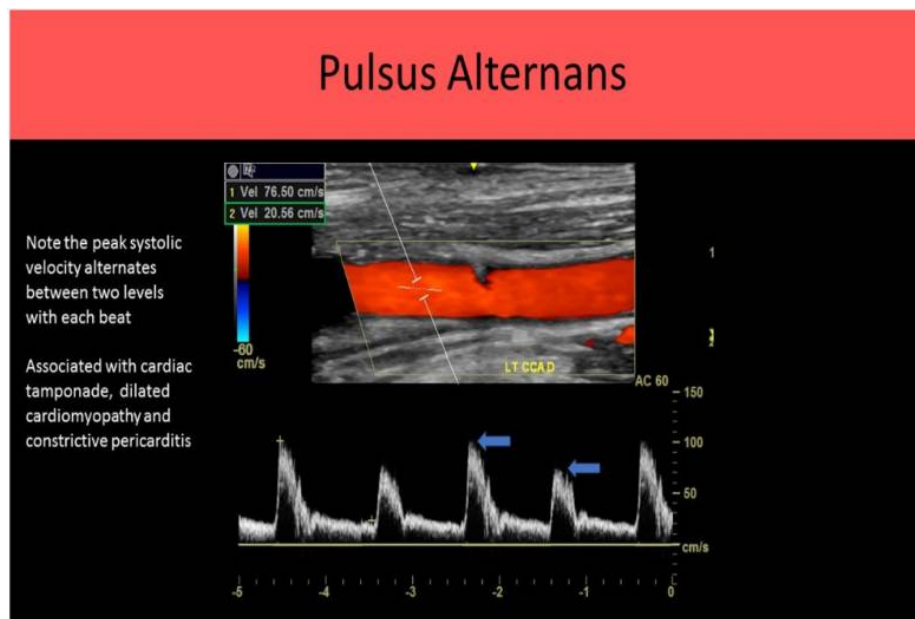


73. What cardiac complications are associated with the findings on the image? view image

- A. mitral stenosis or pulmonary stenosis
- B. coronary artery disease or aortic stenosis
- C. aortic regurgitation or pulmonary regurgitation
- D. cardiac tamponade or constrictive pericarditis

74. The transducer is placed over the mid femoral vein while calf compression is performed. The Doppler waveform demonstrates spontaneous flow with an increase in forward flow with the calf compression. Which of the following describes the findings?

- A. The flow response is normal for the femoral vein.
- B. The popliteal and distal femoral veins have incompetent valves.
- C. The popliteal vein has incompetent valves.
- D. Heart disease is most likely present which causes the abnormal flow response.



75. When preparing for a patient with suspected thoracic outlet syndrome, what other information should you review for pertinent information?

- A. Lab results for hematocrit levels
- B. Chest x-ray
- C. CT of the brain
- D. Aortogram

76. Blood within the pulmonary veins:

- A. contains a high oxygen content and flows into the right atrium
- B. contains a low oxygen content and flows into the left atrium
- C. contains high oxygen content and flows into the left atrium
- D. is normally high resistance

77. When you change the angle on the Doppler cursor, the _____ also changes.

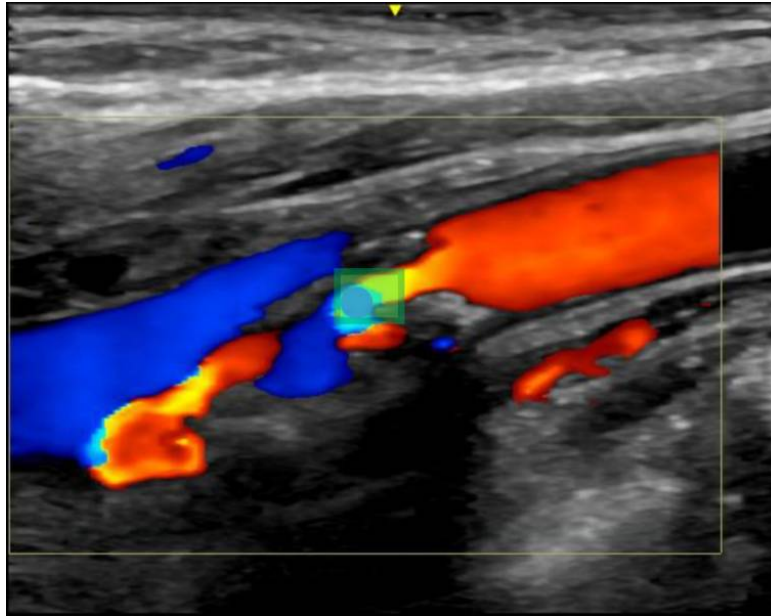
- A. speed of sound in tissue
- B. amplitude of the reflected signal
- C. the volume of flow interrogated
- D. flow velocity displayed

78. Which of the following structures is supplied by branches of the internal carotid artery?

- A. lateral nose only
- B. parietal lobe only

- C. eyes and lateral nose, but not the parietal lobe
- D. eyes, lateral nose and parietal lobe

79. Select the area of the color display where the highest systolic velocity would be demonstrated on PW Doppler evaluation. Click to set the cursor and confirm your answer.



80. Where should the Doppler sample be placed to obtain the highest velocity when evaluating laminar flow?

- A. in the center of the vessel
- B. varies with the type of vessel evaluated
- C. adjacent to the anterior wall
- D. adjacent to the posterior wall

81. Vertigo is a common symptom related to disease of which of the following vessels?

- A. anterior cerebral artery
- B. middle cerebral artery
- C. internal carotid artery
- D. basilar artery

82. A patient presents with splenomegaly and multiple, small, tortuous vessels are visualized in the porta hepatis area of the liver. These vessels most likely represent:

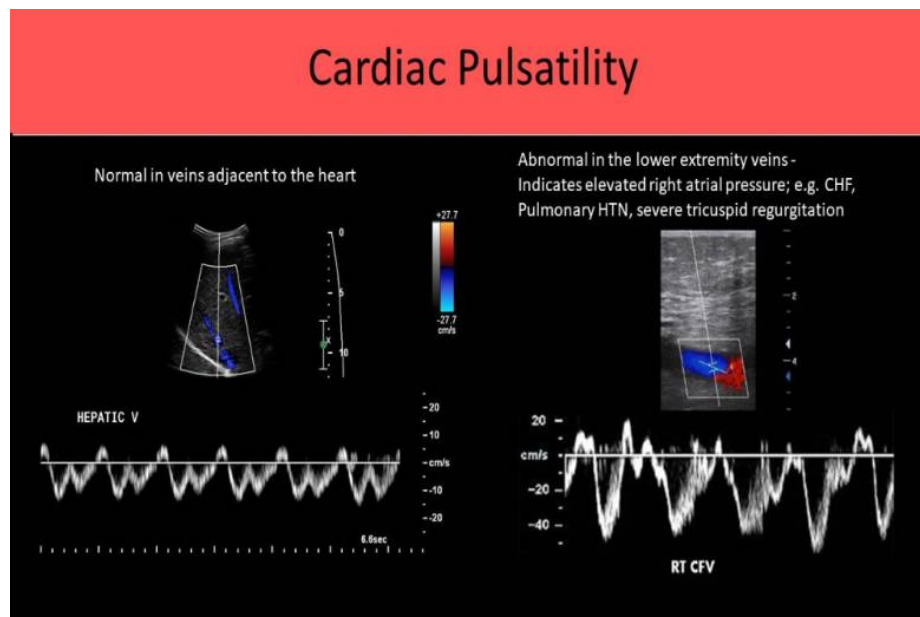
- A. Duplicated hepatic artery
- B. Cavernous transformation
- C. Dilated extrahepatic biliary system
- D. Portal aneurysm

83. Which of the following Doppler findings would indicate significant venous hypertension?

- A. Phasic flow patterns in the lower extremity deep veins
- B. Absence of flow in the deep veins on deep expiration
- C. Increased systolic velocities with decreased diastolic velocities
- D. Triphasic flow patterns in the lower extremity deep veins

84. Anticoagulant therapy is a contraindication for:

- A. ultrasound guided thrombin injection
- B. a transcranial duplex exam
- C. a segmental pressure exam
- D. cold immersion testing



85. A patient presents with gangrene of the foot and toes. The most likely cause of these symptoms is:

- A. extensive deep and superficial vein thrombosis
- B. a 5cm aortic aneurysm with mural thrombus
- C. chronic occlusion of the popliteal artery
- D. coarctation of the aorta

86. Which of the following techniques is used to monitor flow during a fem-pop bypass procedure?

- A. segmental pressures
- B. duplex ultrasound

- C. pulse volume recording
- D. B mode ultrasound

87. If a liver transplant patient has a piggyback anastomosis, how does this affect your evaluation?

- A. there will be a single anastomosis site in the IVC that must be evaluated
- B. there will be two anastomosis sites in the main portal vein that must be evaluated
- C. the hepatic artery anastomosis will not be able to be visualized because it is located deep within the liver tissue
- D. each hepatic vein is connected to the native IVC separately, so there will be three anastomosis sites in the IVC that must be evaluated

88. A critical ICA stenosis will have what effect on the ipsilateral middle cerebral artery flow?

- A. increased systolic flow
- B. increased diastolic flow
- C. no effect
- D. decreased diastolic flow

89. Which of the following vessels typically supplies blood to a carotid body tumor?

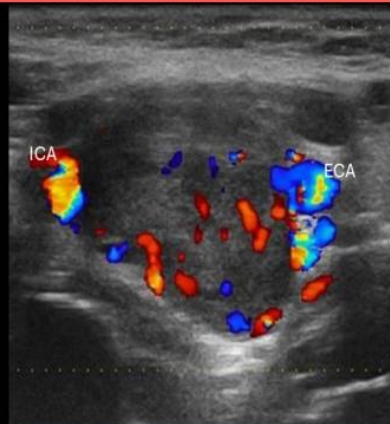
- A. branches of the internal carotid artery
- B. branches of the external carotid artery
- C. branches of the middle cerebral artery
- D. branches of the subclavian artery

90. Letter B indicates which of the following structures?

- A. Internal thoracic artery
- B. Superficial temporal artery
- C. Thyrocervical trunk
- D. Vertebral artery

Carotid Body Tumor

A carotid body tumor typically forms at the CCA bifurcation, between the ECA and ICA. It is a highly vascular tumor and branches of the ECA typically supply the tumor with blood.



Courtesy of Taco Geertma, MD
Ultrasoundcases.info



A minimally invasive technique for closing an iatrogenic subclavian artery cannulation using the Angio-Seal closure device: two case reports.
Szkup PL - J Med Case Rep (2012)

91. D-dimer levels are most accurate for predicting:

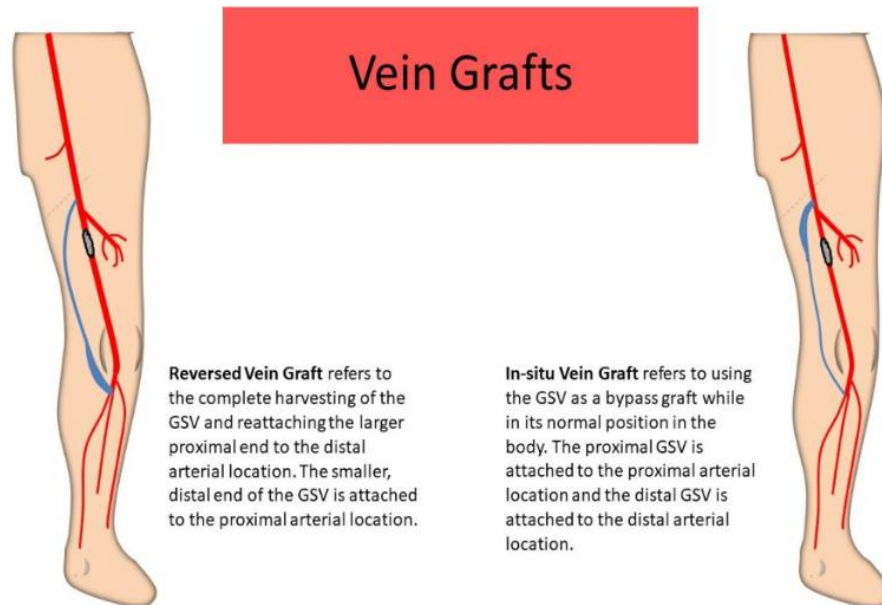
- A. the risk of pulmonary embolism
- B. the absence of DVT
- C. the risk of clot formation
- D. the presence of DVT

92. Flow distal to a significant stenosis will exhibit which of the following characteristics?

- A. higher resistance flow
- B. lower resistance flow
- C. increased velocity
- D. increased resistive index

93. Which of the following correctly describes a reversed vein graft?

- A. the patient must wait at least 48 hrs after the valvulotome procedure is performed to complete the connection of the graft connection
- B. commonly used to correct bilateral iliac stenosis
- C. there is less scarring with a reversed vein graft procedure than an in situ graft procedure
- D. stenosis at the proximal anastomosis is a common complication



94. In the circle of Willis, if a difference _____ is present between the velocities taken in the same vessel on both sides, this can indicate intracerebral disease.

- A. >20%
- B. >30%
- C. >40%
- D. >50%

95. A patient presents complaining of pain and discoloration in their hands when they are cold, as seen in the image. The expected characteristics of the digital PPG waveform in this patient after their hands have been submerged in 40-degree water for 1 minute would include:

- A. Increased amplitude with a prominent dicrotic notch
- B. Triphasic waveform
- C. Monophasic damped waveform
- D. Due to varied levels of vasoconstriction, flow will fluctuate between low resistance and high resistance until hand temperature returns to normal.

96. A 76yr old female with diabetes, HTN, and COPD is sent to the vascular lab for a six month follow up on a carotid stenosis. What is the best patient position to use for the exam?

- A. Seated upright on a stool beside the machine, facing you
- B. Trendelenburg position
- C. Semi-erect with or without a pillow
- D. Supine, neck slightly extended



97. Respiration and _____ are responsible for venous blood flow.

- A. vessel recoil
- B. Valsalva maneuvers
- C. the calf muscle pump
- D. the tunica media layer of the wall

98. What is the most common location of an intracranial aneurysm?

- A. Posterior communicating artery
- B. Internal carotid artery siphon
- C. Middle cerebral artery
- D. Anterior communicating artery

99. Virchow's triad of symptoms includes blood stasis, recent trauma and:

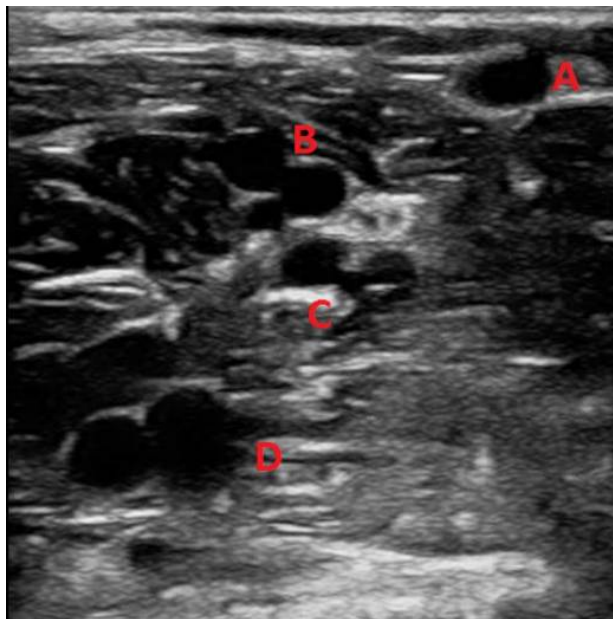
- A. Anemia
- B. Hyperpigmentation
- C. Polycythemia vera
- D. Hyperhydration

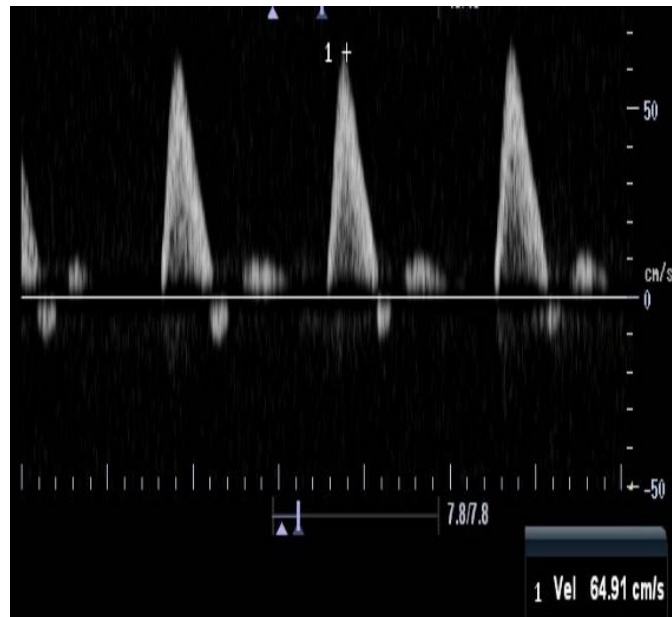
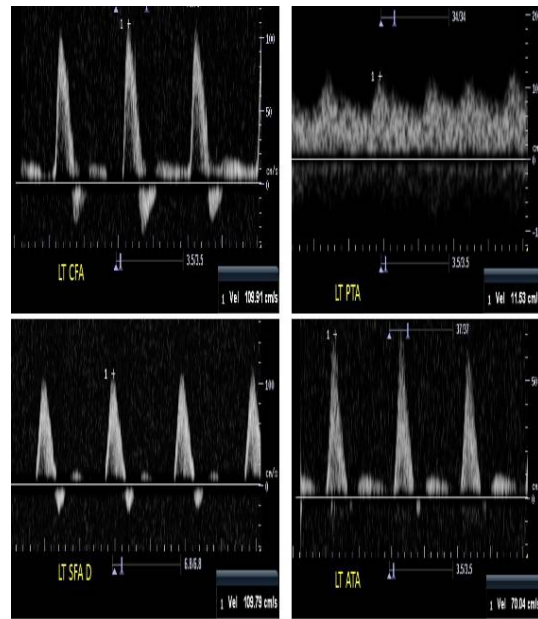
100. If the image displayed is a transverse view of the posterior right lower leg, Letter D correspond to:

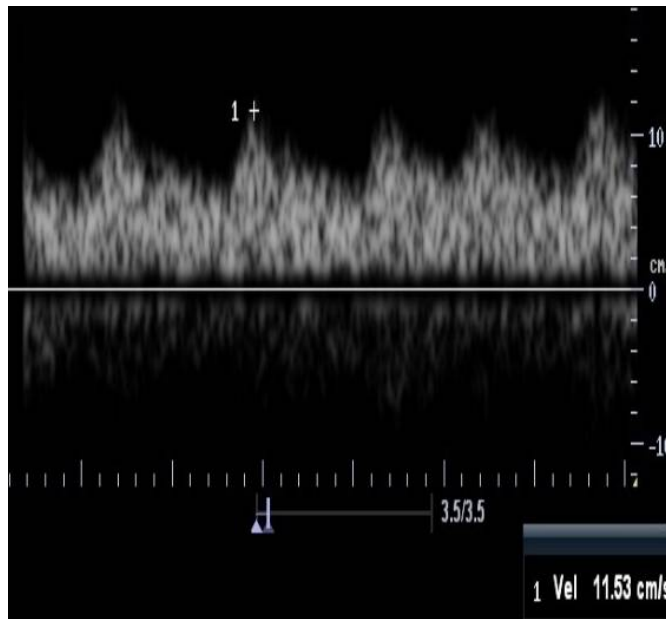
- A. medial gastrocnemius veins
- B. lateral gastrocnemius veins
- C. popliteal vein
- D. posterior tibial veins

101. The images below are from a lower extremity arterial duplex examination. Review the information on the images and choose the waveform below that is most consistent with the expected waveform for the left popliteal artery. Image A Image B Image C Image D

- A. Image D
- B. Image C
- C. Image A
- D. Image B

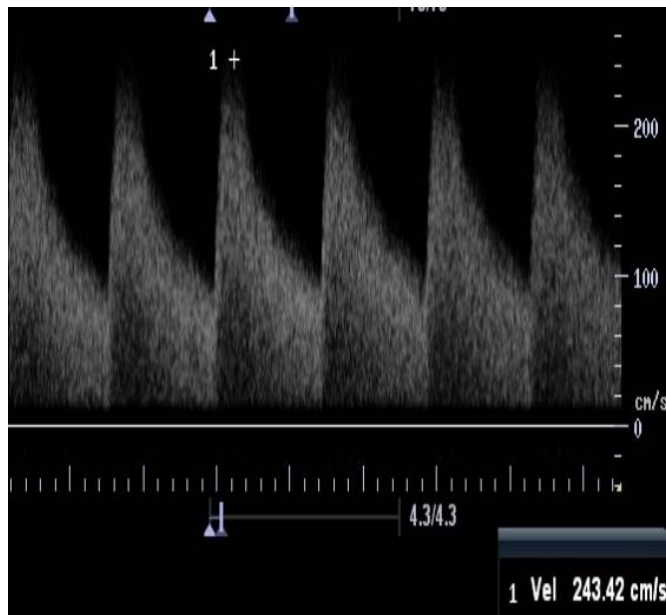


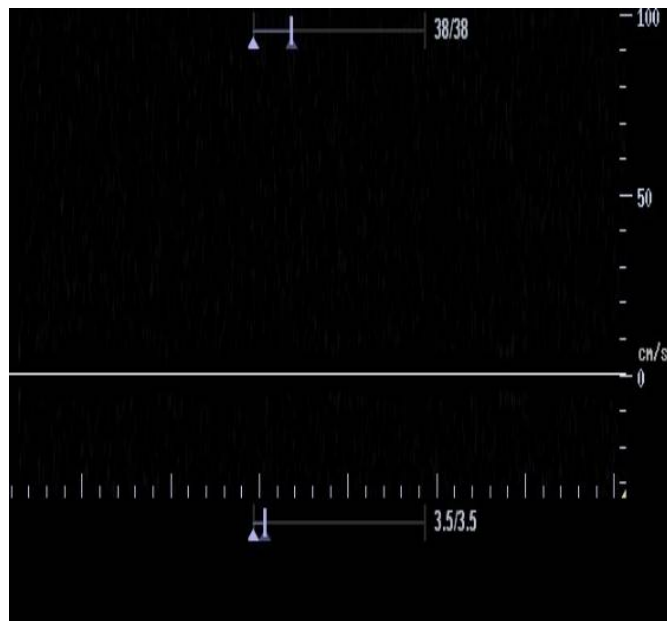




102. Which blood vessel is most commonly evaluated with Doppler to obtain the thigh pressure when performing a segmental pressure exam?

- A. distal calf artery with highest ankle pressure
- B. femoral artery just proximal to the thigh cuff
- C. popliteal artery just above the calf cuff
- D. distal calf artery with lowest ankle pressure





103. Which of the following is not a characteristic of a high-grade stenosis?

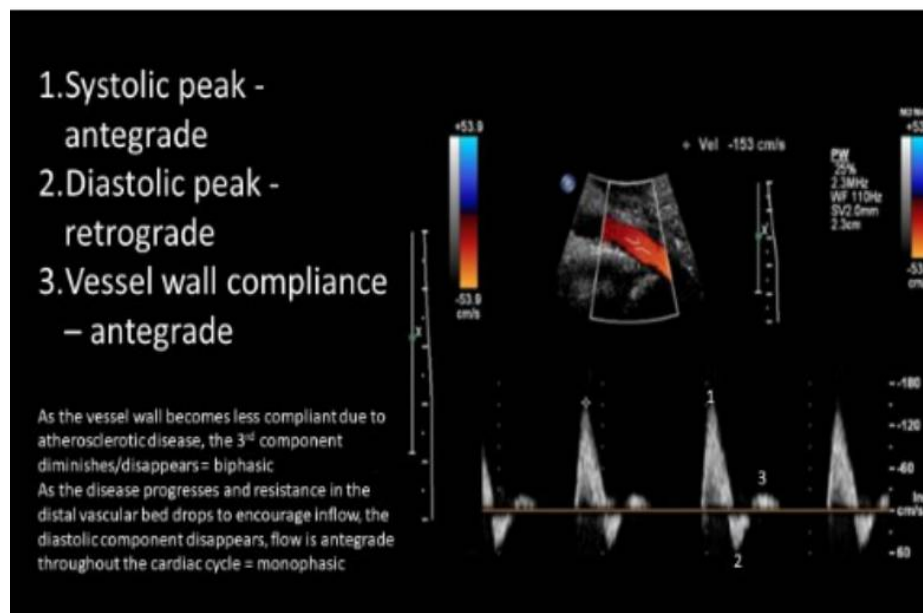
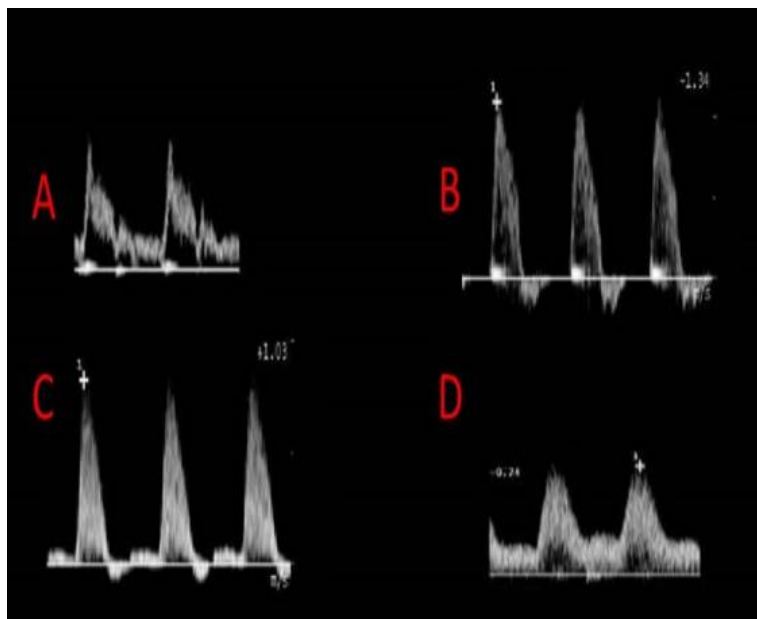
- A. Loss of pulsatility of waveform distally
- B. Laminar flow just distal to the site of stenosis
- C. Spectral broadening just distal to the site of stenosis
- D. Extremely high velocity signals at the site of stenosis

104. Which of the waveforms demonstrates antegrade flow in diastole?

- A. C only
- B. B only
- C. A, C, D
- D. A, B, D

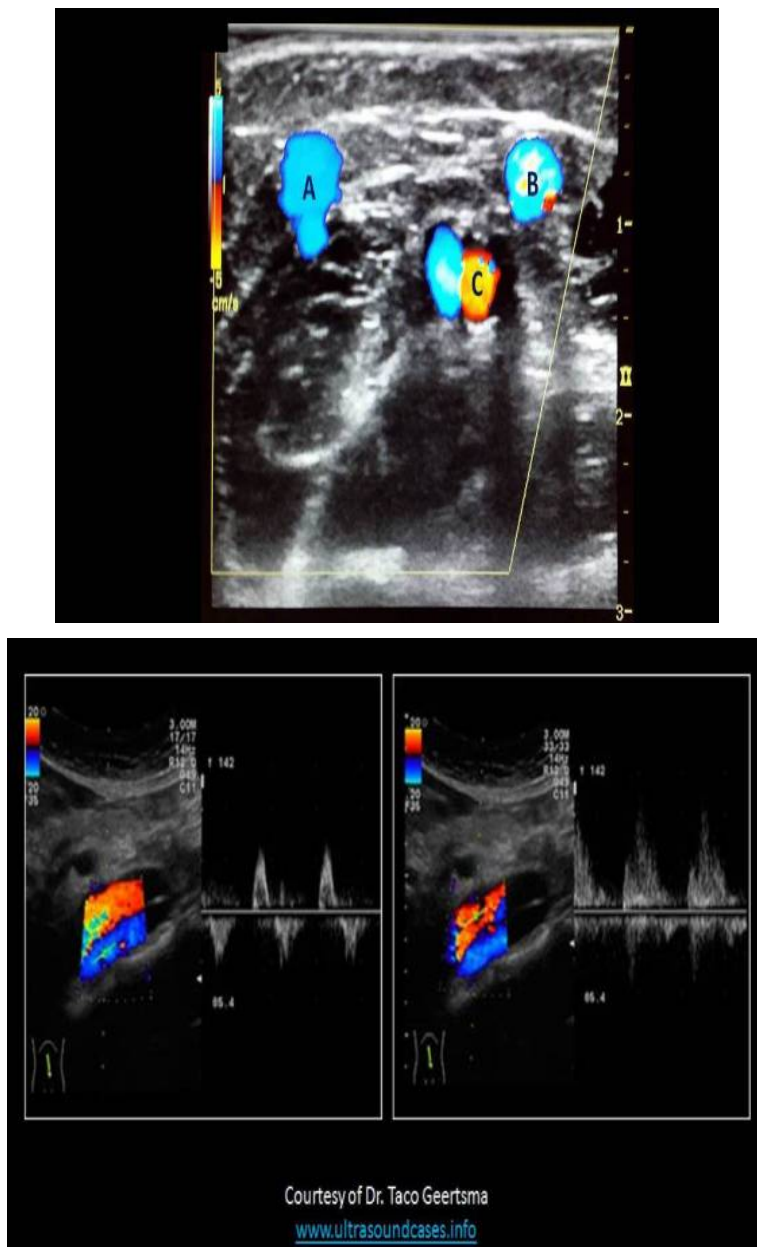
105. The image is a transverse view of the right antecubital fossa. What vessel is indicated by letter C?

- A. cephalic vein
- B. brachial artery
- C. basilic vein
- D. brachial vein



106. The images displayed are from an aortic Doppler evaluation on an ER trauma patient. What should you do upon completion of the exam?

- A.** Provide a preliminary report describing normal flow in the aorta and IVC.
- B.** Leave a preliminary report with the nurse describing the AV fistula and send the exam to the PACS system for a stat interpretation by the radiologist.
- C.** Leave a preliminary report with the nurse describing the mild atherosclerosis in the distal aorta but an otherwise normal exam.
- D.** Leave a preliminary report with the nurse describing the dissection and send the exam to the PACS system for a stat interpretation by the radiologist.



107. Which of the following is related to chronic DVT?

- A. Wet ulceration
- B. Engorged, non-compressible vein
- C. Dry ulceration
- D. Swelling and erythema

108. Which of the following causes of thrombosis is associated with anatomic variations of the thoracic outlet?

- A. indwelling catheters
- B. Paget - Schroetter Syndrome
- C. radiation therapy treatments to the chest

D. shoulder surgery

109. Which of the questions listed below would provide pertinent history information for a lower extremity venous Doppler exam?

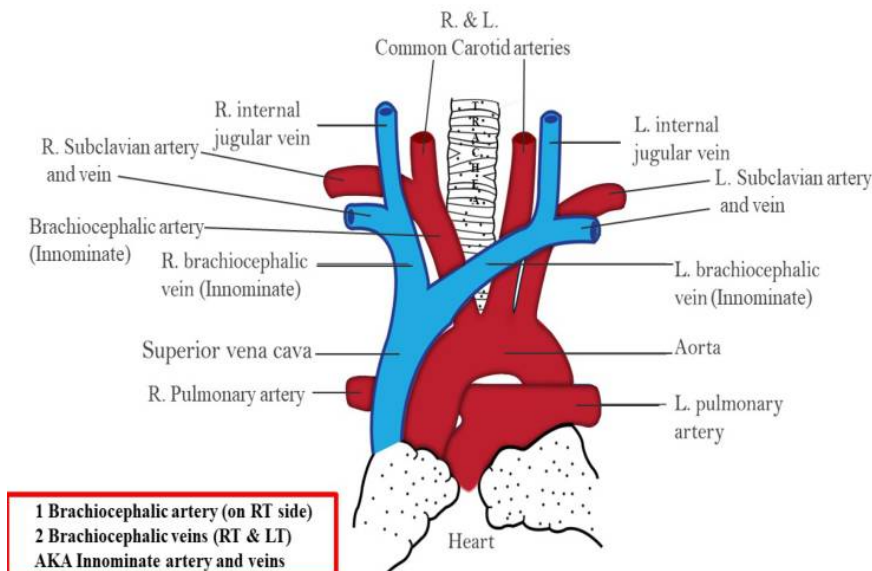
- A. Do your legs/calves hurt more when walking, but the pain stops if you sit and rest?
- B. If you on the edge of the bed with your feet on the floor, does this reduce the swelling?
- C. Does your job require you to stand for long periods of time?
- D. Do you have any ulcer formation on your toes?

110. Which of the following is a limitation of the 2D evaluation of the extremity veins?

- A. location of the brachial vein
- B. increased amount of fibrous tissues and ligaments in the adductor canal
- C. inability to visualize the vessels immediately posterior to the inguinal ligament
- D. patient cannot lay flat for exam

111. The innominate veins course _____ to the innominate artery.

- A. posterior and lateral
- B. inferior and posterior
- C. posterior and medial
- D. anterior and lateral



112. The images are from a Doppler evaluation of the kidneys. What do you expect to find when you evaluate the main renal arteries?

- A. both vessels will demonstrate normal velocities

B. left renal artery will demonstrate flow velocities $>200\text{cm/s}$ and the right renal artery will demonstrate normal velocities

C. both renal arteries will demonstrate stenosis

D. right renal artery will demonstrate flow velocities $>200\text{cm/s}$ and the left renal artery will demonstrate normal velocities

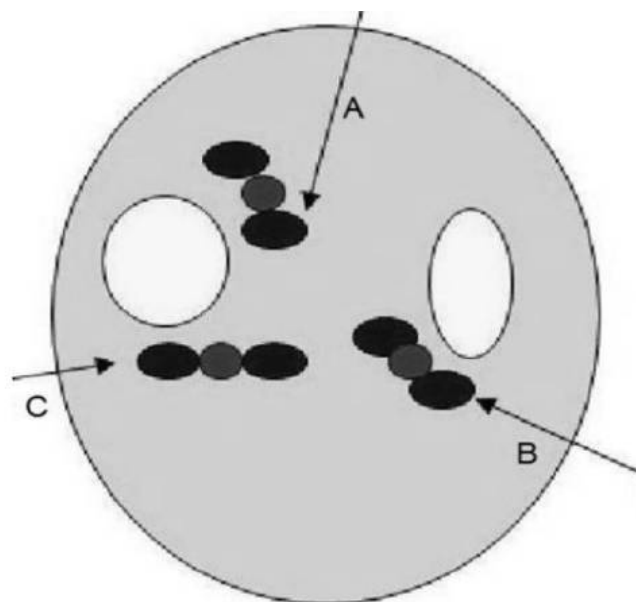
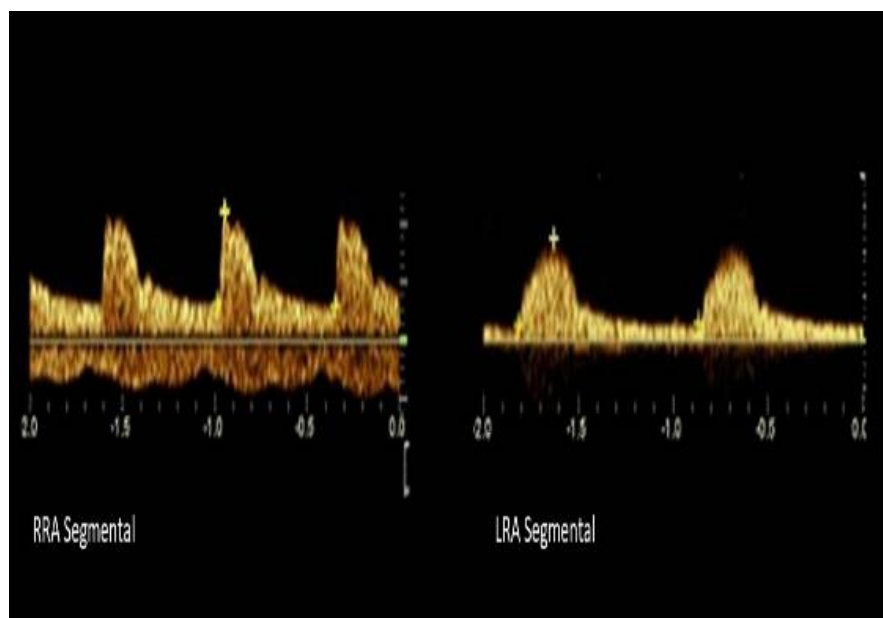
113. What blood vessels are located at letter C?

A. 1 PTA and 2 PTV

B. 2 peroneal arteries and 1 peroneal vein

C. 2 peroneal veins and 1 peroneal artery

D. 1 ATA and 2 ATV



114. Normally, what changes occur in the waveform of the SMA postprandially?

- A. resistance decreases
- B. resistance increases
- C. diastolic flow decreases
- D. resistance and diastolic flow decreases

115. What is the most accurate method of thigh pressure assessment?

- A. Doppler velocity used in the Bernoulli equation to calculate thigh pressure
- B. 4-cu segmental pressure exam
- C. 3-cu segmental pressure exam
- D. Intravenous catheter measures pressure in the superficial femoral vein

116. Which of the following describes the proper technique for evaluating the MCA through the temporal window using TCD?

- A. Sharp anterior angulation with 65mm depth
- B. Midline probe placement, slight anterior angulation with 70mm depth
- C. Sharp anterior probe angulation, 50mm depth
- D. Midline probe placement, slight anterior angulation with 50mm depth

Transtemporal Window

Intracranial Artery	Average Depth Setting	Probe Angulation	Time Averaged Maximum Velocity	Direction of Flow
MCA (M1)	30 - 60mm	Anterior	55cm/s ± 12cm/s	Toward
ACA	60 - 80mm	Anterior	50cm/s ± 11cm/s	Away
PCA (P1)	60 - 70mm	Posterior and Caudal	39cm/s ± 10cm/s	Toward
PCA (P2)	60 - 70mm	Posterior	40cm/s ± 10cm/s	Away
Distal ICA	55 - 65mm	Anterior and Caudal	39cm/s ± 9 cm/s	Bi-directional

117. Hypothenar hammer syndrome and thoracic outlet syndrome are what type of vascular disorder?

- A. Compression syndromes
- B. Atherosclerotic diseases
- C. Fibromuscular syndromes
- D. Autoimmune diseases

118. Which of the following statements about the IVC is correct?

- A. The IVC should normally maintain a relatively constant diameter with respiration.
- B. Congestive heart failure and pulmonary HTN can cause IVC dilatation.
- C. The IVC empties blood into the left atrium.
- D. The IVC is the most posterior vessel in the abdomen.

119. Which of the following is a contraindication for harvesting of the great saphenous vein?

- A. chronic obstruction of the femoral vein
- B. Baker's cyst in the ipsilateral popliteal fossa
- C. valvular incompetence in the great saphenous vein
- D. chronic obstruction of the peroneal veins

120. Which of the following is used to determine the capability of healing a wound or identify a site for amputation?

- A. Sclerotherapy
- B. Photoplethysmography
- C. Transcutaneous oximetry
- D. Ascending venography

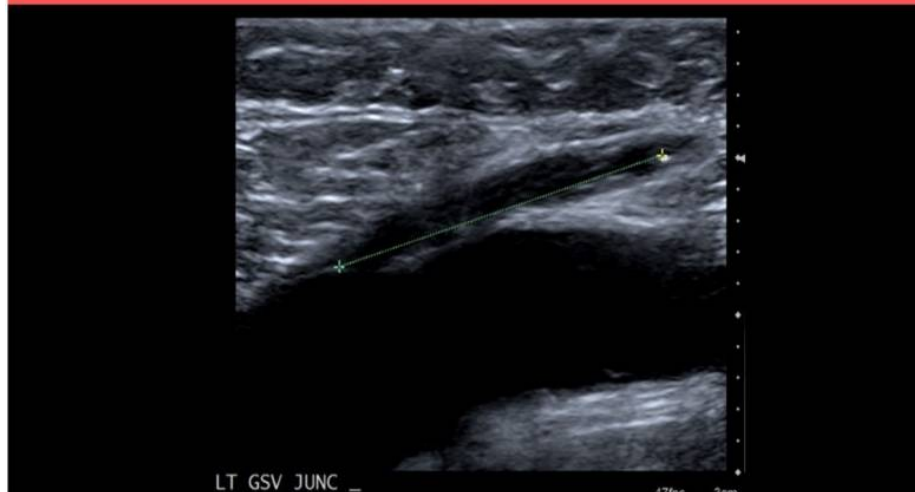
121. Which of the following is a critical finding?

- A. AV fistula with a volume flow rate of 800 ml/min
- B. DVT with a free-floating tail
- C. Atherosclerotic occlusion of the superficial femoral artery
- D. 4cm aortic aneurysm

122. Why is the distance from the most proximal point of the contracted GSV to the saphenofemoral junction measured in a post-ablation exam?

- A. to document the lack of thrombus propagation into the deep system
- B. to assess the length of the segment that still requires treatment
- C. to document that the procedure was unsuccessful
- D. to assess the risk of future reflux in that area

Post-Ablation Assessment

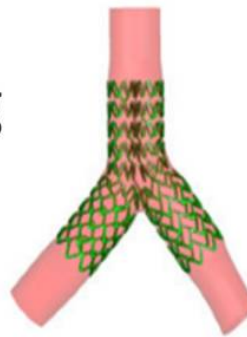


123. If both common iliac arteries demonstrate significant stenosis, what type of treatment will most likely be used?

- A. right axillary - femoral bypass graft
- B. kissing stents
- C. fem-fem bypass graft
- D. left axillary - femoral bypass graft

Arterial Stenting

Kissing
Stent



124. A patient presents for a lower extremity venous Doppler exam due to a history of May-Thurner Syndrome. What vessel should you closely evaluate for related findings?

- A. bilateral distal femoral veins
- B. left iliac vein

- C. bilateral popliteal veins
- D. right iliac vein

125. Which of the following is an advantage of intraoperative vascular ultrasound compared to intraoperative angiography?

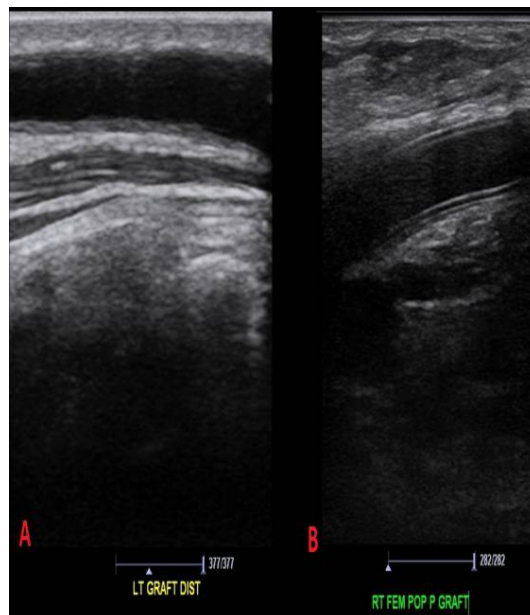
- A. Can be performed during the surgical procedure
- B. Can assess flow in all branches of the intracerebral vessels
- C. Technique does not require vessel puncture to obtain flow information
- D. Provides real time flow evaluation

126. Overgaining on an US exam can cause _____.

- A. overestimation of Doppler velocity
- B. increased posterior shadowing
- C. underestimation of arterial stenosis
- D. blood to appear anechoic

127. Image A is _____. Image B is _____.

- A. Image A: in situ vein graft; Image B: reversed vein graft
- B. Image A: venous graft; Image B: PTFE graft



128. Which of the following correctly explains why a surgeon would perform an intravascular ultrasound immediately after stent placement in small artery?

- A. To evaluate the stent position
- B. To assess the velocity of flow in the stent

- C. To assess the velocity of flow in the artery
- D. To confirm that all plaque was removed from the artery

129. When evaluating the left arm for potential placement of a central venous catheter, you identify multiple small venous collaterals surrounding the subclavian vein. What should you do next?

- A. evaluate the extremity arteries instead for potential catheterization
- B. evaluate the venous system in the right arm
- C. measure the largest collateral for potential pathway for the catheter
- D. tell the physician the patient is not a candidate for catheter placement

130. Which of the following correctly describes a screening exam?

- A. testing that leads to the initial diagnosis
- B. testing used to monitor known disease progression
- C. testing used to determine the presence or absence of disease in an asymptomatic patient
- D. the initial exam following vascular intervention/surgery that is usually done within first month after the procedure

131. Which of the following indicates a significant SMA stenosis?

- A. dilated inferior mesenteric artery that is easily visualized sonographically
- B. pain in the abdomen when hungry that is relieved by eating/digestion
- C. PSV >200cm/s
- D. mesenteric - aortic ratio > 2.0

132. A patient complains of right leg pain and color changes in the leg. When he lies down, the entire right leg becomes pale. When he stands up, the entire right leg becomes reddened. The left leg is asymptomatic. These findings are most suggestive of:

- A. deep venous reflux
- B. right popliteal stenosis
- C. significant right iliac stenosis
- D. significant aortic obstruction

133. Which of the following is a major limitation of a lower extremity segmental pressure exam when compared to a Doppler evaluation?

- A. The segmental pressure technique is only valid for evaluating obstructive disease below the knee.
- B. The segmental pressure technique cannot differentiate external iliac from common femoral stenosis/occlusion.
- C. The segmental pressure technique is only valid for evaluating obstructive disease above the knee.
- D. The segmental pressure technique cannot evaluate obstructive disease in the feet.

134. An AV malformation is most commonly caused by:

- A. hemodialysis access
- B. a congenital defect
- C. trauma
- D. recent angiography procedure

135. Which of the following is important information to record on the final report for a carotid Doppler exam on a patient with 50-69% ICA stenosis?

- A. Doppler angles used on the exam
- B. Mechanical index settings
- C. Transducer frequency used for the exam
- D. Recommendation for TCD exam for associated intracerebral disease

136. Which transducer should be selected for a renal artery Doppler examination for a patient that is 5'5" and 150lbs?

- A. 3.5MHz curvilinear array
- B. 2.5MHz curvilinear array
- C. 3.5MHz linear array
- D. 3.5MHz phased array

137. A clot in the gastrocnemius veins will first propagate into what vessel(s)?

- A. peroneal veins
- B. popliteal vein
- C. deep femoral vein
- D. anterior tibial veins

138. Which of the following symptoms is the most commonly seen with peripheral arterial disease?

- A. Dry ulcerations
- B. Phlegmasia dolens
- C. Claudication
- D. Absent pedal pulses

139. If the left proximal CCA demonstrates a PSV of 35cm/s and the right proximal CCA demonstrates a PSV of 115cm/s, which of the following could explain the findings?

- A. decreased hematocrit levels
- B. congestive heart failure

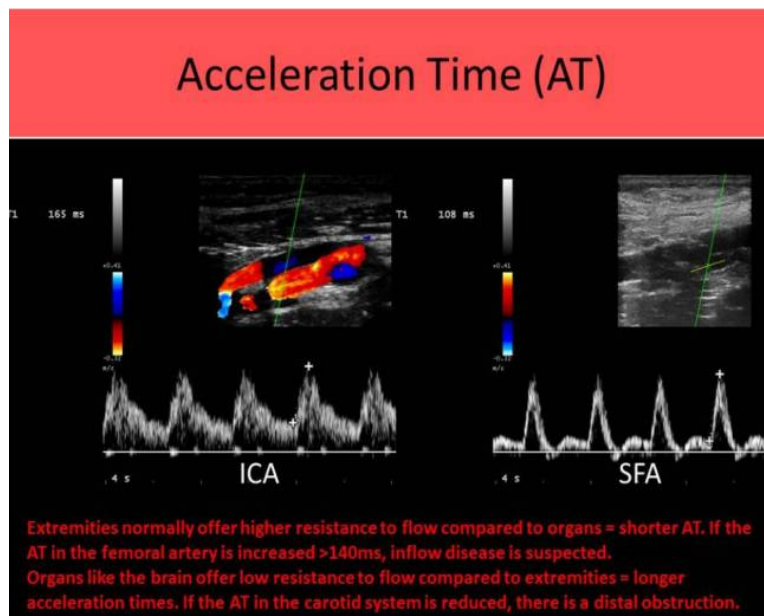
- C. left ICA occlusion
- D. aortic valve stenosis

140. A 30yr old male smoker is referred for an arterial evaluation for pain and pallor of both forearms and hands. The bilateral radial and ulnar arteries demonstrate diffuse, homogeneous intimal thickening and monophasic flow. The arterial lumen characteristics and flow in the upper arms and shoulders are normal. These findings are most consistent with:

- A. Buerger Disease
- B. Medial calcification
- C. Takayasu Arteritis
- D. Atherosclerosis causing tunnel stenosis

141. _____ is defined as the time from onset of systole to the time peak systole occurs.

- A. acceleration time
- B. spatial pulse length
- C. velocity time interval
- D. resistive index



142. A _____ cu is usually used to assess wrist pressure and a _____ cu is usually used to assess digit pressure.

- A. 7cm, 2.5cm
- B. 2.5cm, 1cm
- C. 10cm, 7cm
- D. 12cm, 2.5cm

143. Which cerebral vessel is most commonly involved in an acute cerebral ischemia?

- A. basilar artery
- B. Posterior cerebral artery
- C. Anterior cerebral artery
- D. Middle cerebral artery

144. While scanning the popliteal artery, you notice a thrombus in the popliteal vein. Which vein(s) should be evaluated next for ascending propagation of the clot?

- A. distal femoral vein
- B. ATA and tibioperoneal trunk
- C. GSV
- D. common femoral vein

145. The pre and post exercise pressure information displayed from the right leg demonstrates which of the following?

- A. Single level stenosis
- B. Multiple level stenosis
- C. Raynaud's Syndrome
- D. Normal study

Exercise Pressure Measurement											
	Rest	1	2	3	4	5	6	7	8	9	10
R Ankle (PT):	149	88	125	130	158						
L Ankle (DP):	170	98	131	138	162						
Brachial:	175	194	193	191	191						

146. When performing a PVR exam of the lower extremity arteries, how are the waveforms obtained?

- A. Starting at the right thigh cuff and moving toward the right ankle cuff, then repeating the process with the left leg
- B. Starting at the ankle on the right leg, proceed up through the right thigh before repeating the process with the left leg

- C. Starting at both ankles and moving up through the cu■s on both legs simultaneously
- D. Starting at both thighs and moving down through the cu■s on both legs simultaneously

147. If a renal artery Doppler exam demonstrates normal flow, but stenosis is identified on angiography evaluation, the US results are described as:

- A. true positive
- B. false positive
- C. false negative
- D. true negative

148. You are performing a carotid duplex on a patient that had a carotid endarterectomy 2 weeks ago. There is homogeneous material lining the ICA and bulb and Doppler demonstrates turbulence and elevated velocities. Which of the following is the most likely cause for these findings?

- A. progression of atherosclerotic disease
- B. neointimal hyperplasia
- C. intimal flap and dissection
- D. residual blockage from suboptimal surgical techniques

149. A patient presents with a history of recurrent upper extremity DVT due to Paget Schroetter syndrome. The referring physician ordered a PVR exam of the upper extremities instead of a venous Doppler exam of the upper extremities. Why?

- A. to rule out associated atherosclerotic disease
- B. to rule out Raynaud's syndrome
- C. to rule out thoracic outlet syndrome
- D. to rule out a subclavian steal

150. Abnormal results for a penile Doppler evaluation that suggest impotence include _____ peak systolic flow velocity and a peak pressure _____.

- A. <30cm/s, <60mmHg
- B. <60cm/s, >60mmHg
- C. >60cm/s, >30mmHg
- D. >30cm/s, >60mmHg

151. If a patient complains of a single 12hr episode of right arm paresthesia and paralysis, what will you report on your technologist worksheet?

- A. history of cerebrovascular accident
- B. history of reversible ischemic neurological defect
- C. history of transient ischemic attack

D. history of amaurosis fugax

152. Which of the following indicates a balloon angioplasty of an 80% stenosis of the superficial femoral artery was a success?

- A. decrease in the pulsatility index distal to the treated segment
- B. increased damping in the inflow artery
- C. velocity ratio greater than 2.0
- D. an increase in the ABI of at least 0.15

153. When a patient performs exercise on a treadmill what will be the normal change in the ankle brachial indices?

- A. unchanged or mild increase
- B. unchanged or mild decrease
- C. decreased by less than 0.3
- D. increased by more than 0.3

154. According to the findings on this arteriogram, which of the following would NOT be an expected symptom related to the findings?

- A. Left leg paresthesia
- B. Right leg paralysis
- C. Visual disturbances of the left eye
- D. Left arm paresthesia



155. An upper extremity venous Doppler that is performed for suspected DVT should be performed with the patient in the _____ position.

- A. supine
- B. standing
- C. Trendelenburg
- D. prone

156. A patient complains of left arm numbness and pain that increases in severity when it is cold outside. An upper extremity arterial duplex exam demonstrates a >50% stenosis in the distal subclavian artery. The PPG exam demonstrates an additional decrease in digital flow with cold exposure. What explains these findings?

- A. Secondary Raynaud Phenomenon
- B. Buerger Disease
- C. Primary Raynaud Syndrome
- D. Thoracic Outlet Syndrome

157. The image displayed is from the PTA from an arterial Duplex exam with normal ABIs and minimal atherosclerosis. Which of the following explains the findings?

- A. The wall filter is set too high and is removing pertinent information from the display.
- B. The waveform is normal and is consistent with the other findings of the exam.
- C. The Doppler gain is too low and pertinent information is not displayed.
- D. Medial calcification is causing increased resistance to the flow causing a peaked appearance of the monophasic waveform.

158. Which of the following is not proper procedure for a venous PPG evaluation of the legs?

- A. Always place the sensor over the largest varicose vein in the calf
- B. A tourniquet is applied to the leg after initial testing shows a VRT < 20 sec
- C. The optimal speed for the strip chart recorder is 5mm/sec
- D. Patient is seated with legs dangling over edge of bed for the exam

159. According to the Appropriate Use Criteria, which of the following is an appropriate indication for a screening ankle-brachial index exam?

- A. 58yr old male with diabetes
- B. Diabetic neuropathy with normal pulses
- C. Patient with low Framingham risk score
- D. Nocturnal pain or cramping with normal pulses on clinical exam

Appropriate Use Criteria – Lower Extremity Arterial Duplex/Physiologic Exam

ALWAYS APPROPRIATE

- Claudication
- Rest pain
- Ulceration or gangrene
- Leg infection with non-palpable pulses
- Clinical signs of embolism (eg blue toes)
- New symptoms in patients with previously normal or abnormal duplex or ABI exam
- 1 month follow up baseline exam post-revascularization
- Follow up exam post-revascularization with new or worsening symptoms
- 6-8 month follow up on stable patients after angioplasty, placement of stent, vein bypass graft, or prosthetic bypass graft
- After first year post-revascularization, annual follow up on stable patients after angioplasty, placement of stent, vein bypass graft, or prosthetic bypass graft
- Any of the following post-intervention symptoms: pulsatile groin mass, thrill or bruit at groin, severe pain or prominent hematoma formation
- Screening ABI exam on patients with femoral bruit or diminished pulses
- Screening ABI exam on patients >50yrs with diabetes or smoking history
- Screening ABI exam on patients >70yrs

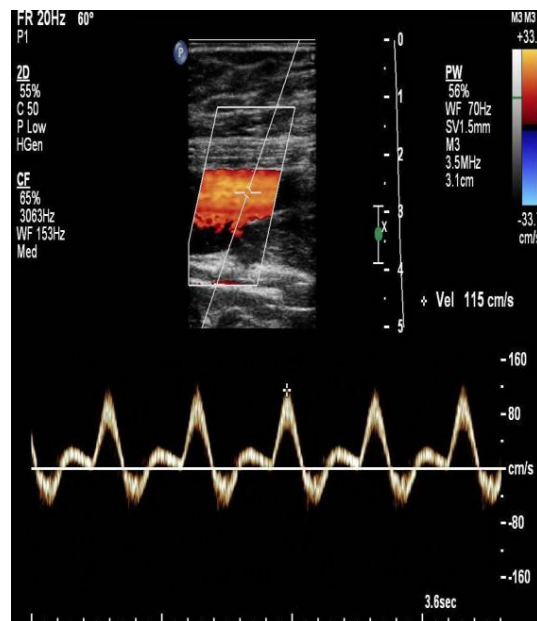
Appropriate Use Criteria – Lower Extremity Arterial Duplex/Physiologic Exam

INAPPROPRIATE

- Nocturnal pain or cramping with normal pulses on clinical exam
- Normal pulses, but no leg/foot hair
- Lower extremity swelling with normal pulses
- Screening exam on a patient with low Framingham risk score
- Diabetic neuropathy with normal pulses
- Follow up exam on stable patients with normal ABI on baseline exam

160. What console controls should be adjusted to improve the image?

- increase the color Doppler scale
- increase the PW Doppler scale
- drop the baseline
- more than one of the above



161. What can be done to improve the visualization of the anterior aortic wall?

- A. decrease the threshold setting.
- B. scan from a more lateral approach
- C. center the probe with the ultrasound beam perpendicular to the aortic wall
- D. increase the compression

162. When Thrombin is injected in a pseudoaneurysm, the clotting should be begin:

- A. within 5-10 seconds post- injection
- B. after 1 minute of manual compression following the injection
- C. within 5-10 minutes post- injection
- D. within 24 hours

163. According to the Appropriate Use Criteria, which of the following is an appropriate indication for an lower extremity venous duplex exam for venous insufficiency?

- A. Hyperpigmentation
- B. Positive D-dimer test
- C. Suspected or diagnosed pulmonary hypertension
- D. Telangiectasia

164. Review the image and the system settings listed along the right side. Use your mouse to position your cursor over the setting that should be adjusted to improve the diagnostic quality of the Doppler tracing. Click to set the cursor and confirm your answer.

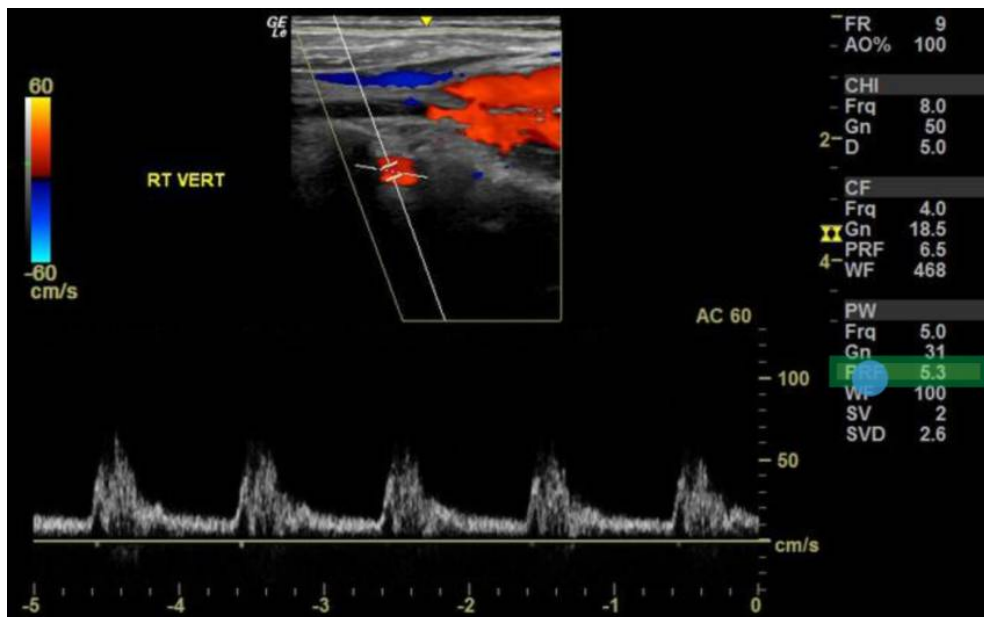
Appropriate Use Criteria – Lower Extremity Venous Duplex Exam (Insufficiency)

ALWAYS APPROPRIATE

- Hyperpigmentation and skin changes seen with venous insufficiency, no visible varicose veins
- Visible varicose veins with associated symptoms (leg heaviness, discoloration, ulceration)
- Active or healed venous ulcer
- Vein mapping for endovenous ablation
- New or worsening symptoms in a patient previously treated with endovenous ablation
- Suspected or diagnosed pulmonary embolism

INAPPROPRIATE

- Telangiectasia (spider veins)
- Prior endovenous saphenous ablation procedure with no residual symptoms



165. Which of the following is an absolute contraindication to an endovenous ablation?

- A. uncooperative patient
- B. current antibiotic use
- C. history of diabetes mellitus
- D. daily aspirin use

166. How do you adjust your technique to reduce or eliminate cross talk on a Doppler tracing?

- A. Use a smaller sample size
- B. Turn down the Doppler gain
- C. Increase the PRF

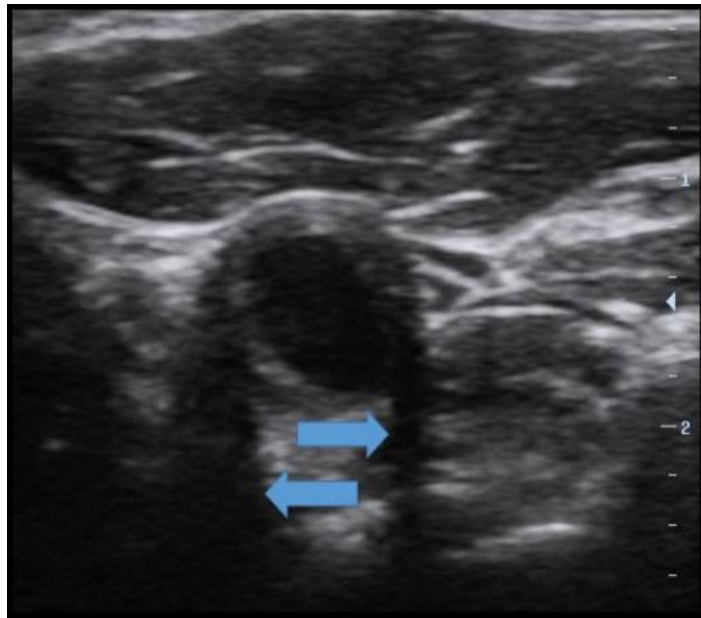
D. Use the same approach and incident angle but use a CW non-imaging probe

167. Which of the following correctly describes the positive predictive value (PPV) of a testing technique?

- A. The ability of a test to detect disease when it IS present.
- B. If the test result is positive, PPV is the probability that the patient actually has the disease
- C. The ability of a test to rule out disease when it IS NOT present.
- D. If the test result is negative, PPV is the probability that the patient does not have disease

168. What 2D artifact is demonstrated on the image?

- A. edge shadowing/defocusing
- B. edge enhancement
- C. side lobes
- D. mirror image



169. Why did the technologist steer the 2D image while scanning during the thrombin injection?

- A. to make the US beam more perpendicular to the long axis of the needle
- B. to make the US beam more parallel to the long axis of the needle
- C. to provide a better view of the posterior enhancement from the pseudoaneurysm
- D. to improve the color flow demonstration



170. Which upper extremity vein is most commonly evaluated for use as an arterial bypass?

- A. median cubital vein
- B. cephalic vein
- C. radial vein
- D. brachial vein

Answer Key

1	B	2	C	3	B	4	B
5	A	6	D	7	A	8	D
9	C	10	C	11	A	12	A
13	B	14	C	15	A	16	B
17	D	18	A	19	A	20	D
21	C	22	B	23	A	24	A
25	C	26	A	27	C	28	C
29	-	30	C	31	A	32	D
33	D	34	B	35	D	36	B
37	B	38	D	39	A	40	B
41	A	42	B	43	D	44	C
45	B	46	C	47	B	48	A
49	D	50	C	51	B	52	C
53	A	54	B	55	C	56	C
57	C	58	C	59	A	60	D
61	D	62	B	63	C	64	C
65	C	66	C	67	D	68	A
69	C	70	D	71	B	72	C
73	D	74	A	75	B	76	C
77	D	78	D	79	-	80	A
81	D	82	B	83	D	84	A
85	A	86	B	87	A	88	D

89	B	90	D	91	B	92	B
93	D	94	B	95	C	96	C
97	C	98	D	99	C	100	C
101	C	102	A	103	B	104	C
105	B	106	D	107	A	108	B
109	C	110	B	111	D	112	B
113	A	114	A	115	C	116	D
117	A	118	B	119	A	120	C
121	B	122	A	123	B	124	B
125	C	126	A	127	B	128	A
129	B	130	C	131	A	132	C
133	B	134	B	135	A	136	A
137	B	138	C	139	C	140	A
141	A	142	A	143	D	144	A
145	A	146	D	147	C	148	D
149	C	150	A	151	C	152	D
153	A	154	A	155	A	156	A
157	A	158	A	159	A	160	D
161	C	162	A	163	A	164	-
165	A	166	B	167	B	168	A
169	A	170	B				